A

PROJECT REPORT ON

"Automated Electricity Metering System"

SUBMITTED TO SAVITRIBAI PHULE PUNE UNIVERISTY FOR THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF

BACHELOR OF ENGINEERING IN ELECTRONICS AND TELECOMMUNICATION

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Academic Year 2024-2025

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CERTIFICATE

This is to certify that the project entitled, **"Automated Electricity Metering System",** has been successfully completed by,

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towards the partial fulfillment of the degree of **Bachelor of Engineering** in **Electronics and Telecommunication** Engineering to be awarded by the **Savitribai Phule Pune University, Pune** at **Pune Vidyarthi Griha's College of Engineering and Technology G. K. Pate (Wani) Institute of Management, Pune** during the academic year 2024-2025.

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ABSTRACT

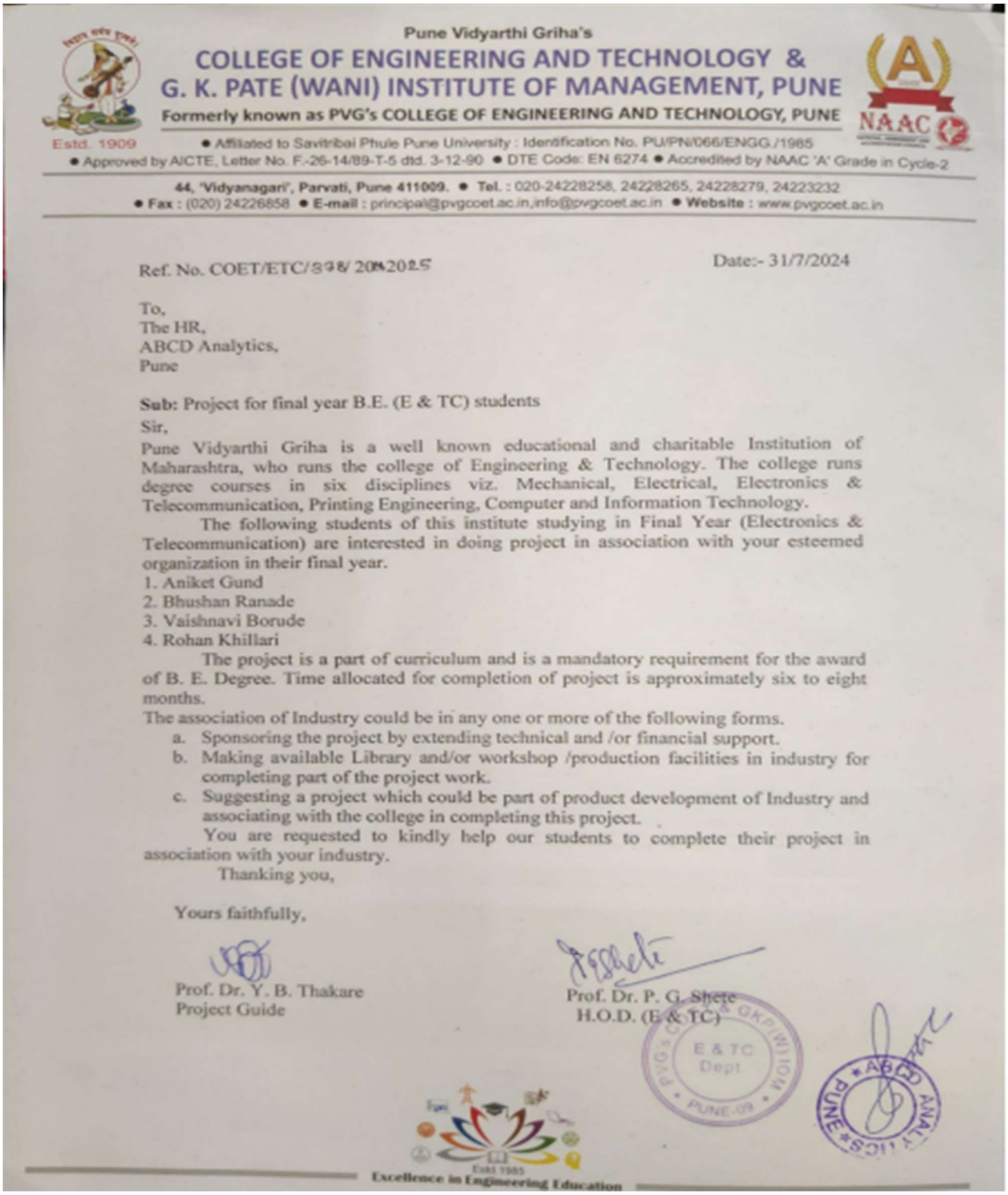
Traditional energy meters suffer from several limitations, making energy monitoring and billing inefficient and less user-friendly for both consumers and utility providers. A significant issue is the lack of real-time data, which leaves consumers unaware of their energy consumption patterns until they receive their bills. This often results in unexpected expenses and limits users' ability to monitor or control their usage effectively. Additionally, traditional meters rely on manual readings, which are time-consuming, prone to errors, and create inefficiencies for utility providers, leading to potential billing inaccuracies.

To address these challenges, this project proposes the development of an Automated Energy Metering System with Prepaid Functionality. This system introduces real-time energy monitoring and a user-friendly prepaid billing model, empowering users to track and manage their energy consumption proactively. The smart meter is built around an energy metering IC that accurately measures critical parameters such as voltage, current, power, and total energy consumption. An ESP32 microcontroller processes this data and transmits it to a remote server via a Wi-Fi and use GPRS module to send SMS on contact number.

The data collected on the server is displayed through interactive dashboards, accessible to both users and utility providers. Users can log in to monitor their energy usage in real-time, receive low-balance alerts, and recharge their accounts seamlessly. This prepaid model enables users to actively manage their energy expenses and consumption. Utility providers benefit from enhanced operational efficiency, gaining insights into usage patterns, remote monitoring capabilities, and the ability to control meters when necessary.

By integrating real-time data collection, automated metering, and a user-centric billing approach, this project offers a practical and modern solution for energy management. It enhances transparency, control, and efficiency for consumers while streamlining billing and monitoring processes for utility providers. Overall, this Automated Energy Metering System represents a significant advancement over traditional systems, leveraging technology to address their limitations and deliver a more effective energy management solution.

Internship Letter :



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To,
The HR,
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Pune

Sub: Project for final year B.E. (E & TC) students

Sir,

Pune Vidyarthi Griha is a well known educational and charitable Institution of Maharashtra, who runs the college of Engineering & Technology. The college runs degree courses in six disciplines viz. Mechanical, Electrical, Electronics & Telecommunication, Printing Engineering, Computer and Information Technology.

The following students of this institute studying in Final Year (Electronics & Telecommunication) are interested in doing project in association with your esteemed organization in their final year.

1. Aniket Gund
2. Bhushan Ranade
3. Vaishnavi Borude
4. Rohan Khillari

The project is a part of curriculum and is a mandatory requirement for the award of B. E. Degree. Time allocated for completion of project is approximately six to eight months.

The association of Industry could be in any one or more of the following forms.

a. Sponsoring the project by extending technical and /or financial support.
b. Making available Library and/or workshop /production facilities in industry for completing part of the project work.
c. Suggesting a project which could be part of product development of Industry and associating with the college in completing this project.

You are requested to kindly help our students to complete their project in association with your industry.

Thanking you,

Yours faithfully,

Prof. Dr. Y. B. Thakare
Project Guide

Prof. Dr. P. G. Shete
H.O.D. (E & TC)

E & TC Dept

Excellence in Engineering Education

LIST OF ACRONYMS

Wi-Fi:	Wireless Fidelity
GPRS:	General Packet Radio Service
LCD:	Liquid Crystal Display
kWh:	kilowatt-hour
AMR:	Automated Meter Reading
LoRa:	Long Range
PF:	Power Factor
AC:	Alternating Current
DC:	Direct Current
CT:	Current Transformer
PT:	Potential Transformer
IoT	Internet of Things
PCB:	Printed Circuit Board
RTC:	Real Time Clock
SMS:	Short Message Service

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Chapter 1
INTRODUCTION

1.1 Introduction:

The Automated Electricity Metering System helps users monitor their electricity usage in real time and manage their bills efficiently. It consists of a PZEM-004T energy meter to measure power consumption and an ESP32 microcontroller to process the data. The collected data is transmitted to a remote server using Wi-Fi, where users and administrators can track it through a dashboard. Additionally, a GPRS module is used to send SMS notifications to users about balance alerts and important updates. The system follows a prepaid model, allowing users to recharge their accounts. If the balance reaches zero, a relay module automatically disconnects the power supply, ensuring proper billing and controlled energy usage.

To enhance security and prevent tampering, the system includes multiple tamper detection mechanisms. A ZMPT voltage sensor detects unusual voltage fluctuations, which may indicate tampering attempts. A Hall effect sensor identifies magnetic field interference, ensuring that external magnetic manipulation does not affect meter readings. Additionally, a switch is attached to the meter case, which remains in a pressed state under normal conditions. If someone tries to open the meter, the switch is released, triggering an alert to notify administrators of possible tampering.

An LCD display provides real-time updates on energy consumption and balance, allowing users to stay informed about their usage. With its real-time monitoring, prepaid billing, tamper detection, and user-friendly interface, the system offers a secure, efficient, and smart solution for managing electricity consumption in residential, commercial, and industrial settings.

1.2 Motivation

It is observed that traditional energy meters have several limitations such as the absence of real-time energy monitoring, lack of transparent and accurate billing, and no efficient way to manage electricity usage. Consumers often face difficulties in tracking their electricity consumption, resulting in unexpected high bills. Utility providers also face issues like manual meter reading, inaccurate billing, difficulty in detecting energy theft, and challenges in effective load management.

1.3 Objectives

- Design smart meter with prepaid billing compliant with relevant standards and security for residential application.
- Implement wireless communication for real-time data transmission.

1.4 Outcomes

- Improves billing accuracy
- Increases transparency in energy usage
- Helps in early detection of billing errors
- Reduces manual work and human errors
- Provides real-time monitoring of electricity consumption
- Helps consumers avoid unexpected high bills
- Supports energy-saving by making users aware of their usage
- Enables easy and secure mobile payments
- Reduces chances of energy theft through tamper detection
- Contributes to better load management for utility providers

1.5 Social relevance of the project

Automated Metering System will transform energy management by significantly improving billing accuracy and customer satisfaction. This advanced metering technology promotes fairness and transparency, ensuring that customers are billed based on precise usage data, which can reduce disputes and build trust between consumers and utility providers. By automating the detection of service issues and minimizing the need for physical disconnections, supporting uninterrupted service for users. Additionally, the system will help in energy conservation efforts by providing real-time data that empowers consumers to make informed choices about their energy consumption. Overall, this project will contribute to a more efficient, and sustainable energy infrastructure that benefits both the community and the environment

Chapter 2
LITERATURE SURVEY

Electricity metering systems are used to measure the amount of electrical energy consumed by a residence, business, or appliance. This measured data allows utilities to bill customers accurately based on their energy consumption over a specific period. Electricity meters operate by continuously measuring the instantaneous voltage and current to give energy used in kWh [1].

The fig 2.1 shows evolution of electricity meters began with electromechanical meters, often referred to as watt-hour meters, which were the standard for many years. These meters measure electricity usage through electromagnetic induction, where a metal disc rotates at a speed proportional to the power consumed. The total energy usage is indicated by the number of disc rotations. Although reliable, electromechanical meters require utility personnel to manually record readings and are limited in accuracy due to their mechanical nature. With technological advancements, digital meters gradually replaced electromechanical meters. With the advancement of digital technology, electronic digital meters emerged, built on digital microtechnology (DMT). These meters replaced the mechanical components of electromechanical meters, offering greater accuracy and efficiency. They laid the foundation for smart meters by introducing remote functionalities such as automated meter reading (AMR) and remote meter reading (RMR), allowing energy suppliers to monitor consumption data from a distance without manual intervention. Additionally, these meters allowed for modular expansion, enabling new features to be added as technology advanced, including load leveling and real-time pricing (RTP) capabilities, which helped suppliers manage energy distribution more efficiently [1].

The most recent advancement in metering technology is the smart meter, which goes beyond AMR by offering two-way communication. Smart meters provide real-time or near real-time data on energy consumption, support remote management, and enable advanced features like time-of-day pricing, power outage notifications, and power quality monitoring. These meters are often part of an Advanced Metering Infrastructure (AMI), a system that enables enhanced energy management and responsiveness for both consumers and utility providers [3].

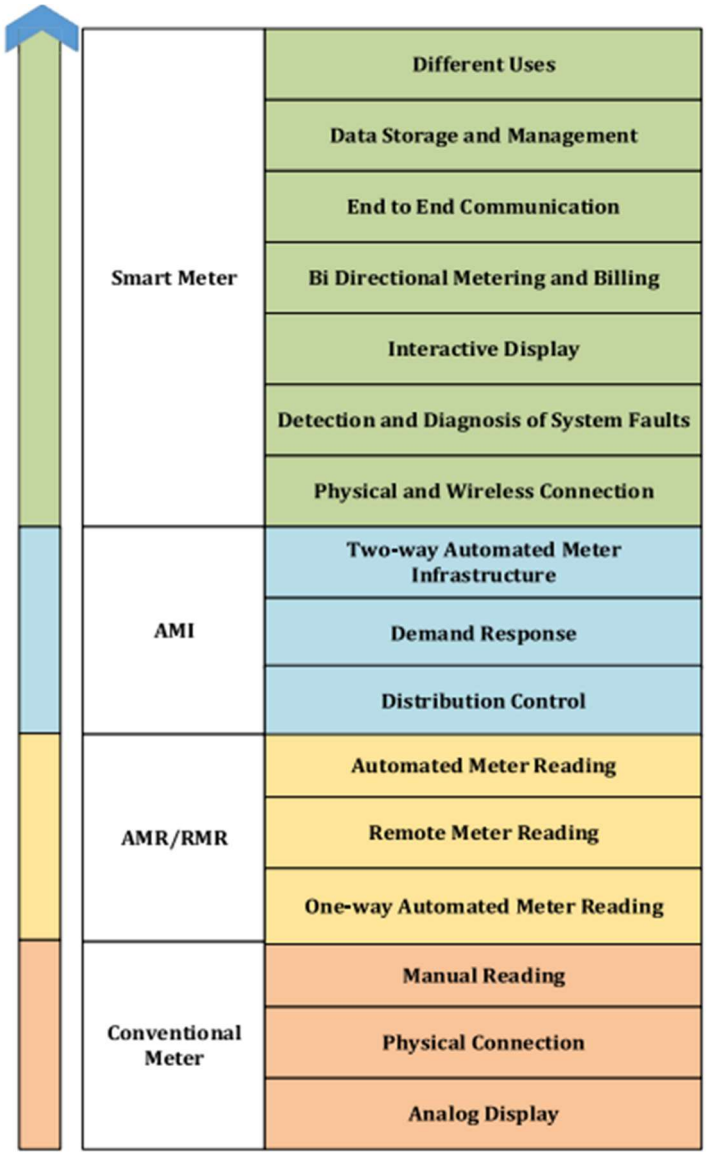


Fig 2.1 Evolution of smart meters [1]

The article titled "A Unified Metering System Deployed for Water and Energy Monitoring in Smart City" discusses the development and implementation of an integrated smart metering system designed to monitor water and energy usage in smart cities, particularly in the context of India. The traditional metering methods for water and energy are often inefficient, time-consuming, and prone to inaccuracies, leading to customer dissatisfaction. The proposed system aims to automate the data collection process, providing real-time information and reducing human error. The system integrates water flow sensors and the PZEM-004T energy meter, transmitting data through Low Power Wide Area Network (LPWAN) technologies, such as LoRa, which is cost-effective and suitable for long-range communication. The system enables utilities to remotely monitor and manage energy and water consumption, providing both the utility provider and customers with real-time data. It also has capabilities to prevent overloading, detect energy theft, and notify utility companies in case of anomalies [2].

Fig 2.2 from Article named “Design, Implementation, and Deployment of an IoT Based Smart Energy Management System” describes the design and implementation of a comprehensive load control and monitoring system for electrical grids, aimed at real-time evaluation of grid efficiency and management of connected appliances. The proposed system, which integrates both hardware and software, enables detailed monitoring and control of loads such as air conditioning and lighting, with the added ability to generate analytical management reports that provide insights into grid performance. The hardware component of the system features an intelligent smart meter (SM) designed using the MSP430F67641 microcontroller, a three-phase energy monitoring device, and a NodeMCU module for bi-directional wireless communication with the Untrack-based management platform. The SM architecture incorporates three key microcontroller system-on-chips (SoCs) for distinct functionalities. The MSP430F67641 MCU, developed by Texas Instruments, measures critical grid parameters, including voltage, current, active, reactive, and apparent power, as well as power factor and frequency, with high precision. The TM4C129x MCU, also from Texas Instruments’ Tiva C series, is based on ARM-Cortex-M architecture and manages data storage and remote control of loads such as cooling and lighting circuits. It also supports environmental sensors for variables like humidity and temperature and connects to the NodeMCU through a UART interface. Finally, the NodeMCU IoT platform handles wireless data transmission to the IoT middleware using the MQTT protocol over IEEE 802.15.4, enabling remote monitoring and control [1].

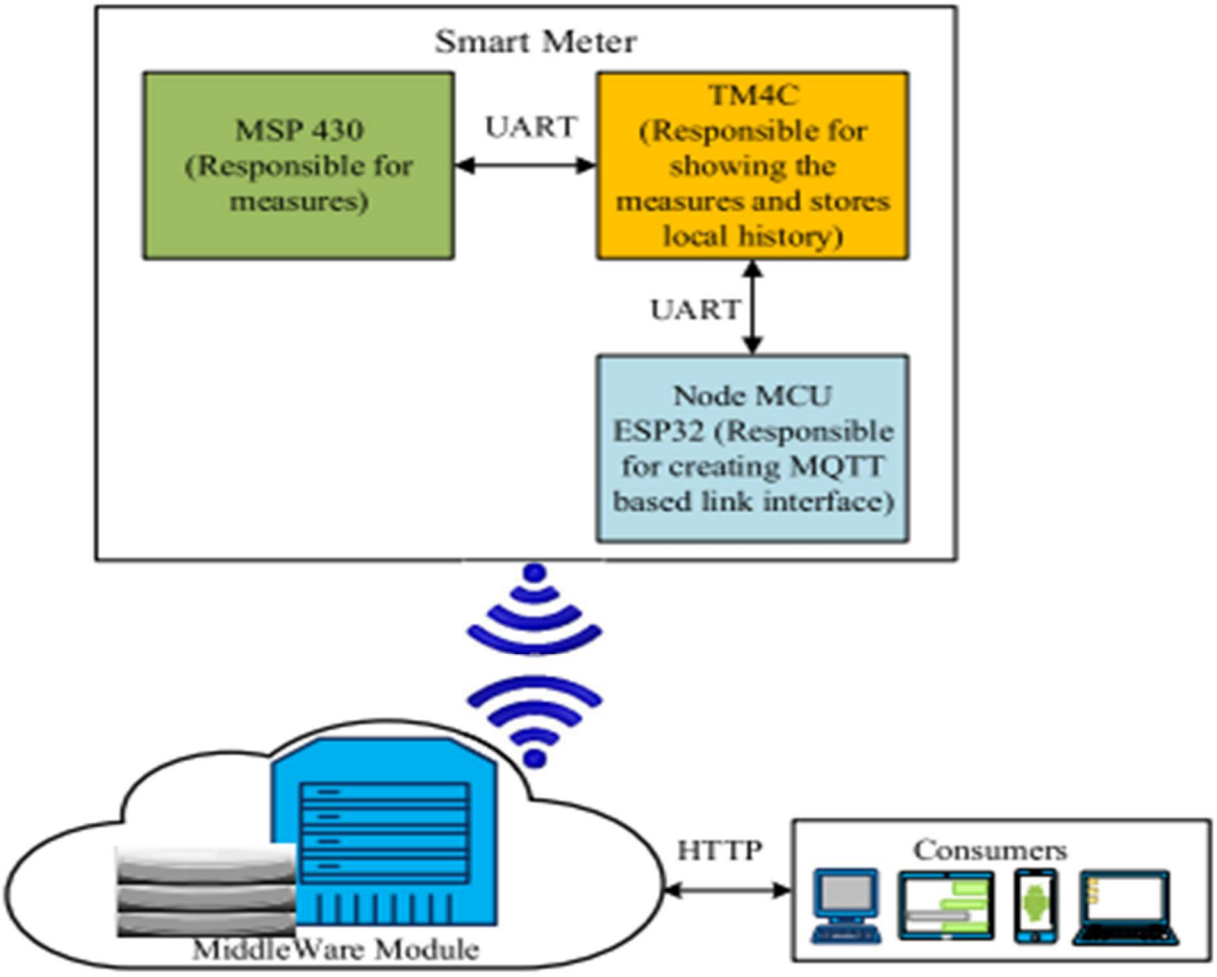


Fig 2.2 Smart Energy Management System [1]

In this paper [3] author presents the design of a Smart Energy Meter (SEM) that integrates multiple components to ensure secure, robust, and resilient energy measurement and management. The SEM is designed with two main computing units: CUA (Arduino-UNO) and CUB (Arduino-WeMos D1), which work in tandem to monitor and control various parameters such as power, voltage, current, and environmental data like temperature and humidity. The SEM also features a tamper detection sensor, GPS module, real-time clock, LoRa WAN module, and a digital multipurpose meter to enhance its functionality and security. The system ensures data integrity and security by isolating the customer's network (HAN) from the utility network (UC). CUA handles the gathering and processing of energy data, encrypting and sending it through a LoRa WAN network to Data Concentrators (DCs), ensuring redundancy and high reliability. The data sent to CUB, which controls IoT devices in the HAN, is updated every 30 seconds, providing real-time feedback to the consumer. The communication layer relies on LoRa WAN for long-range, low-power data transmission with AES encryption, ensuring secure end-to-end communication. The SEM uses two data concentrators (DCA and DCB) for redundancy, transmitting GPS and Received Signal Strength Indicator (RSSI) values to avoid data loss or modification and ensure reliability in case of hardware failures or cyberattacks. To prevent physical and data layer attacks, the design incorporates various protective measures. CUA, for example, checks for tampering and prevents unauthorized access by triggering predefined anti-theft actions. The data layer ensures that the data remains unmodified by using a one-way communication link between CUA and CUB. Furthermore, the system uses encrypted communications via LoRa WAN to guard against attacks such as man-in-the-middle and session hijacking

The computational stack for smart meter analytics is designed to efficiently process the vast amount of data collected by smart meters at different levels of the energy system. At the consumption end, smart meters are kept simple to reduce complexity and cost. However, for larger entities like offices or factories, more sophisticated smart meters with advanced hardware and software resources are employed. The computational workload is distributed across three main layers: edge, fog, and cloud, each handling different tasks based on the complexity and requirements of the application. Edge computing processes data locally at the smart meter level, minimizing latency and bandwidth requirements. While it is suitable for basic tasks like load power quality metering and energy savings, edge devices are limited in computational power and cannot handle more complex analytics. Fog computing serves as an intermediary between edge and cloud, providing more computational resources than edge devices while still being closer to the data source. Fog nodes, controlled by a fog server, handle more advanced metering functionalities, such as load monitoring, theft detection, and fault detection, offering benefits like reduced latency, better security, and privacy. Cloud computing, on the other hand, offers flexible, on-demand computational resources and is ideal for tasks that require large-scale processing, such as neural network training, data consolidation, and visualization. While public clouds are commonly used for general purposes like energy audits and simple metering, private clouds are preferred for sensitive applications, such as power quality measurements and asset management. Cloud computing offers scalability and cost reduction but also introduces challenges such as cybersecurity risks, high infrastructure costs, and potential network congestion [4].

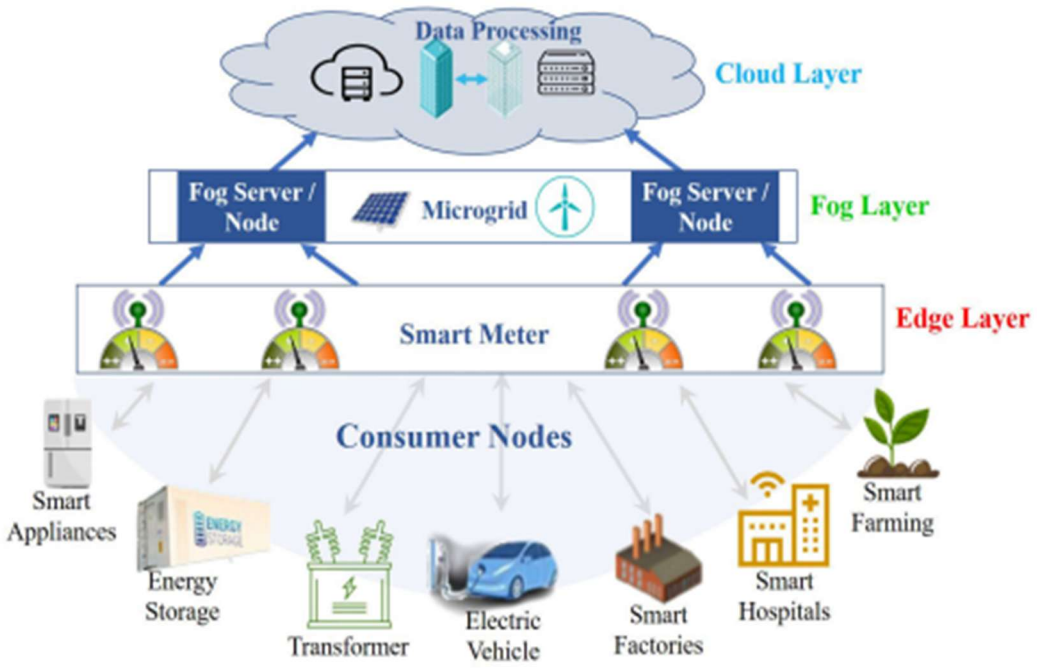


Fig 2.3 Computational stacks for smart meter analytics [4]

Power loss, particularly due to power theft, is a significant issue for distribution networks. Power theft refers to the illegal or unmetered use of electricity, which leads to considerable financial losses for utilities, with an estimated annual loss of 6-10 billion INR in India alone. Various methods of theft include meter bypassing, direct line hooking, and meter tampering, particularly in rural and suburban areas. While numerous power theft detection algorithms exist, many are limited in scope, unable to detect complete bypass or line hooking thefts, and often require complex infrastructure. The advent of smart grids and IoT-based technologies offers a solution to this issue. Smart meters, capable of real-time communication with a central server, can help in monitoring power usage and detecting theft. The proposed system leverages IoT technology for theft detection, allowing for the monitoring of individual consumer usage and detecting any unusual patterns indicative of theft. The system uses a linear regression-based approach for theft detection and is designed to be computationally efficient and cost-effective, making it feasible for widespread implementation. The architecture of the system consists of three layers: the device layer (smart meters), the client layer (consumer and authority apps), and the server layer (cloud server). These layers communicate via APIs using HTTP requests, with the server processing data from the smart meters to identify potential theft and send alerts to authorities. This approach aims to provide real-time monitoring and theft detection in an efficient, scalable manner [5].

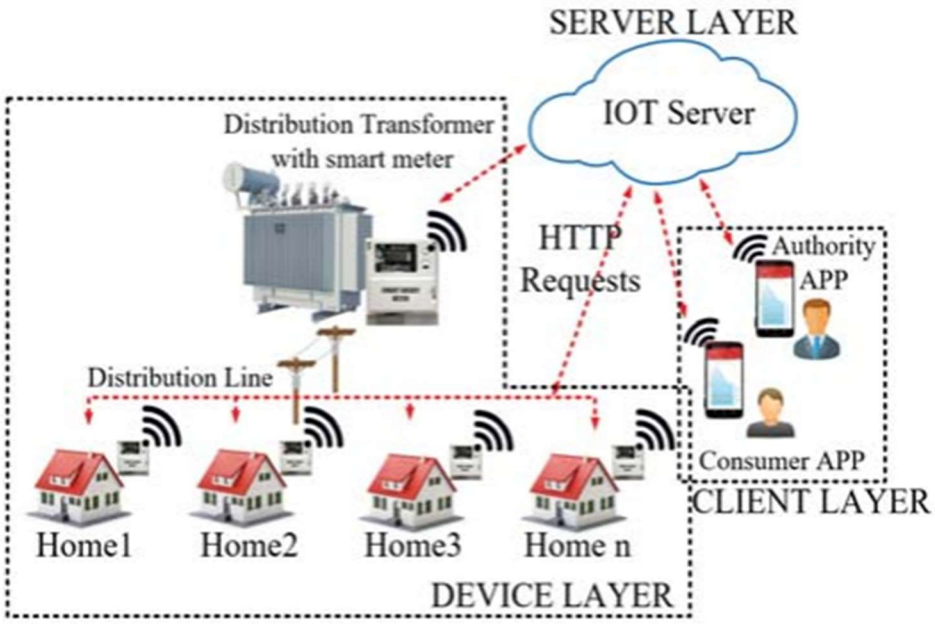


Fig 2.4 layered architecture of smart meter based on power theft detection [5]

In summary, the literature survey demonstrates the progressive transformation of electricity metering systems, driven by the need for accurate billing, energy management, and enhanced operational efficiency. From the early electromechanical meters to modern smart meters, each generation of metering technology has introduced new capabilities to meet evolving demands. Smart meters stand out as a advancement in metering technology, offering real-time data on energy consumption, two-way communication, and remote management. These capabilities not only improve the efficiency of utility operations but also empower consumers to make informed decisions about energy usage. Additionally, smart meters play an essential role in detecting power theft, a significant source of financial losses in many regions, by leveraging real-time monitoring and pattern detection.

However, deploying smart metering systems also poses challenges. Issues such as data security, privacy, and the requirement for reliable communication infrastructure need to be addressed. Innovative solutions, including secure communication protocols like LoRa WAN, data encryption, and resilient system architectures, are being explored to enhance the robustness of these systems against cyber threats and ensure data integrity. Furthermore, the computational stack used in smart metering—encompassing edge, fog, and cloud layers—provides a scalable approach for managing the vast data generated by smart meters. Each layer contributes distinct computational capabilities, from local processing on the edge to extensive analytics in the cloud, optimizing both performance and cost. Balancing these layers is crucial for efficient and scalable energy management.

In summary, the future of electricity metering is strongly oriented towards smart and secure solutions. By advancing smart meter technology and addressing the associated challenges.

Chapter 3
DESIGN OF THE PROPOSED SYSTEM

3.1 System Introduction

The Automated Electricity Metering System is designed to efficiently manage energy consumption using a modular architecture. The system starts with an AC input, which powers both the load and the metering module. This module measures key electrical parameters such as voltage, current, power, and power factor. The collected data is processed by an ESP32 microcontroller, which formats it for display and remote monitoring.

For data transmission, the system uses Wi-Fi to send real-time energy consumption data to a remote server, where the utility provider records and analyzes it for billing. Additionally, a GPRS module is integrated to send SMS alerts to users about their energy usage, balance updates, and notifications.

A prepaid payment system allows users to recharge their accounts and monitor energy consumption in real-time. If the balance reaches zero, a relay module automatically disconnects the power supply, ensuring controlled energy usage. A 16x2 LCD display provides real-time information on consumption, balance, and other details, making the system user-friendly.

To enhance security, the system includes tamper detection to prevent unauthorized access. A Real-Time Clock (RTC) module ensures accurate timestamps for precise billing and consumption logs.

The system is scalable and flexible, allowing future upgrades such as IoT integration, advanced data analytics, and support for renewable energy sources. It can be used in residential, commercial, and industrial applications, providing an efficient, automated, and user-controlled energy management solution.

3.2 Meter Calibration and MSCB standards for residential Single-Phase AC meter:

The Maharashtra State Electricity Distribution Company Limited (MSEDCL, commonly called MSCB or MSEB) follows the IS 13779 standard for residential single-phase AC static energy meters.

Key points from IS 13779 and MSEDCL tender specs:

- Accuracy Class: Typically, Class 1.0 or Class 2.0 (±1% or ±2% error allowed).
- Voltage Range: 230V ±20%
- Current Range: 0–30A or 10–60A (as per meter rating)
- Frequency: 50Hz ±5%

3.3 Electrical Parameters to be measured by the System:

The Automated Electricity Metering System employs the PZEM-004T module to monitor several key electrical parameters essential for accurate energy management and billing. These parameters include:

1. Voltage (V):
   Voltage represents the electrical potential difference supplied to the load. Accurate voltage measurement (typically 80–260V AC) is crucial for monitoring supply quality and detecting abnormal conditions that may indicate faults or tampering and is measured in volts (V).

2. Current (A):
   Current indicates the flow of electric charge consumed by the connected load. Measuring current (up to 10A or 100A depending on the module version) allows the system to determine how much electricity is being drawn by appliances or the entire premises at any moment. This is fundamental for real-time consumption tracking and is measured in ampere(A).

3. Active Power (W):
   Active power is the actual power consumed by electrical devices to perform useful work, such as lighting, heating, or running motors. It is calculated as the product of voltage, current, and power factor. Monitoring active power helps users understand their real-time energy usage and optimize load management and is measured in watts (W).

4. Active Energy (kWh):
   Active energy is the cumulative amount of electrical energy consumed over time, measured in kilowatt-hours (kWh). This parameter forms the basis for electricity billing, as utilities charge consumers based on the total energy used within a billing period and is measured in kilo-watt hour(kWh).

5. Power Factor (PF):
   Power factor is the ratio of active power to apparent power and indicates how efficiently electrical power is being converted into useful work. A low power factor can result in higher losses and increased charges from utilities. Monitoring power factor helps identify inefficiencies and potential corrective actions.

6. Frequency (Hz):
   Frequency measures the number of cycles per second of the AC supply, typically around 50–60 Hz. Stable frequency is essential for the proper operation of electrical equipment. Deviations can signal supply issues or instability in the grid and is measured in hertz (Hz).

3.3.1 Power Calculations and Relationships:

While the PZEM-004T module directly measures active power (P) and active energy (kWh), the system calculates reactive power (Q) and apparent power (S) using the following relationships derived from the measured parameters:

a) Reactive Power (Q):
   Reactive power represents the energy oscillating between the source and reactive components (e.g., inductors, capacitors) in the load. It is calculated using the formula:

   1. $Q = V \times I \times \sin(\phi)$

where ϕ is the phase angle between voltage and current, derived from the power factor ($\cos(\phi)$) measured in volt-amperes (VA).

b) Apparent Power (S):
Apparent power is the vector sum of active and reactive power, representing the total power supplied to the circuit:

$$1. \quad S=V\times I$$

It is used to size electrical infrastructure (e.g., wires, transformers) and assess load capacity and measured in volt-amperes reactive (VAR).

c) Power Triangle:
The relationship between active, reactive, and apparent power is visualized using the power triangle (Fig. X), where:

Active power (P) is the horizontal component.

Reactive power (Q) is the vertical component.

Apparent power (S) is the hypotenuse.

$$S=\sqrt{P2+Q2},\cos(\phi)=SP$$

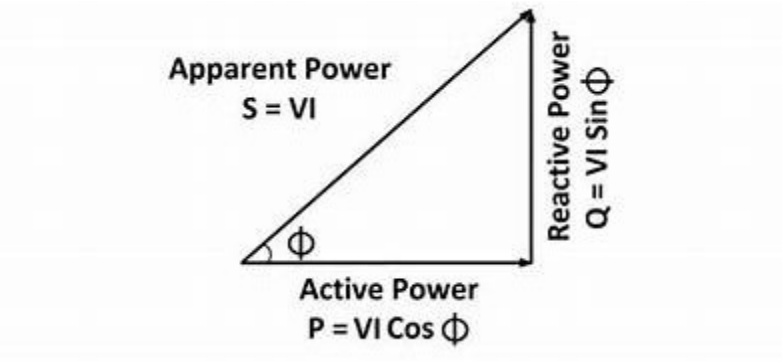


Fig.3.1 Power Triangle

3.4 System Requirements

3.4.1 Hardware Requirements:

A. Functional Requirements:

- Real-Time Energy Monitoring: The system must measure and display key electrical parameters in real-time
- Prepaid Billing Functionality: The system must support a prepaid billing model, allowing users to recharge their accounts and monitor their balance in real-time.
- Remote Data Transmission: The system must transmit energy consumption data to a remote server at regular intervals
- Tamper Detection and Security: The system must include tamper detection mechanisms to identify unauthorized access or tampering. In case of tampering, the system must send an alert to the utility provider and log the event for further action.
- User and Utility Dashboards: The system must provide an interactive dashboard for users to monitor their energy consumption, balance, and recharge options. The utility provider must have access to a dashboard for monitoring multiple meters, analyzing usage patterns, and managing billing.

B. Technical Requirements:

Meter Design:

- Voltage Range: 220–250V AC (single-phase).
- Current Range: 0–30A
- Frequency: 50 Hz
- Power Factor Range: 0–1, with an accuracy of ±1%.
- Power Measurement Range: 0–7.5 kW

Power Supply Requirements:

- Input Voltage: 220–250V AC (single-phase).
- Output Voltage: 5V DC
- Current Output: 4A
- Power Consumption: Approximately 10W for the entire meter unit.

Microcontroller Requirements:

- Processing Capability: Must support real-time data processing and communication with external modules.
- Memory: Sufficient RAM and Flash memory to store and process energy data.
- Connectivity: Must support communication protocols such as UART, I2C, and SPI for interfacing with sensors, displays, and communication modules.

Communication Module Requirements:

- wireless communication for data transmission to a remote server.
- Data Rate: Must support a minimum data rate of 40 kbps for reliable transmission.
- Power Consumption: Must have low power consumption in sleep mode (e.g., <1 mA) to ensure energy efficiency.

Display and User Interface Requirements:

- Display Type: Must include an LCD or similar display for real-time energy data visualization.
- Display Size: Must support at least 20 characters per line and 4 lines for clear readability.
- Power Consumption: Must operate at 5V DC with low current consumption (e.g., <25 mA with backlight).

Relay Module Requirements:

- Coil Voltage: Must operate at 5V DC.
- Contact Rating: Must support at least 10A at 250V AC for reliable power disconnection.
- Switching Time: Must have a fast response time (e.g., <10ms) to ensure quick disconnection when the balance reaches zero.

Tamper Detection Requirements:

- Sensor Type: Must include a tamper detection mechanism (e.g., Hall Effect sensor or Reed switch).
- Sensitivity: Must detect even subtle tampering attempts with high precision.
- Response Time: Must have a fast response time (e.g., <10µs) to ensure quick detection and alert generation.

Real-Time Clock (RTC) Requirements:

- Accuracy: Must provide accurate timekeeping with minimal drift (e.g., ±2 minutes/year).
- Battery Backup: Must include a battery backup (e.g., CR2032) to maintain timekeeping during power outages.
- Interface: Must support I2C or similar communication protocols for easy integration with the microcontroller
-

3.4.2 Software Requirements:

- Functional Requirements:

  User Dashboard:

  - User authentication and profile management.
  - Real-time electricity consumption tracking.
  - Display of daily, weekly, and monthly usage in graphical format.
  - Recharging functionality.

- Billing history access.
- Alerts for low balance , tampering or unusual usage.
- Login and Logout functionality.

Admin Dashboard:

- View total users, active energy meters, and pending alerts.
- Manage users (view, update, and control relay status).
- View billing history for all users.
- Monitor alerts related to low balance and tampering.
- Analyse peak consumption hours and average monthly consumption.
- Update admin profile and access support.
- Login and Logout functionality.

3.5 System Flow Chart

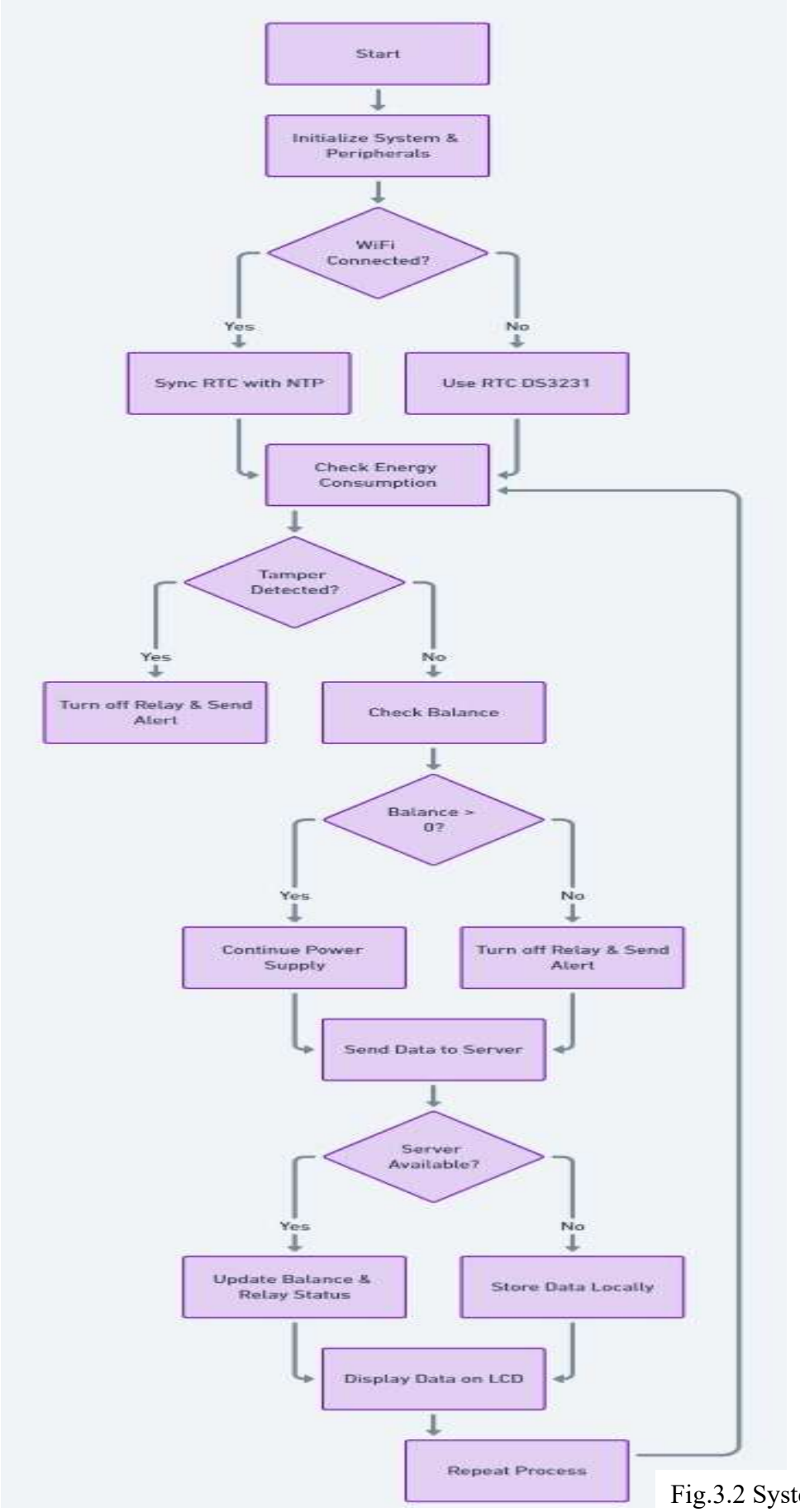


Fig.3.2 System Flowchart

3.6 Block Diagrams

The system architecture is divided into two main parts: the consumer-side and the utility-side. The consumer-side focuses on energy measurement, data processing, and user interaction, while the utility-side handles data storage, billing, and remote management.

A. Consumer side:

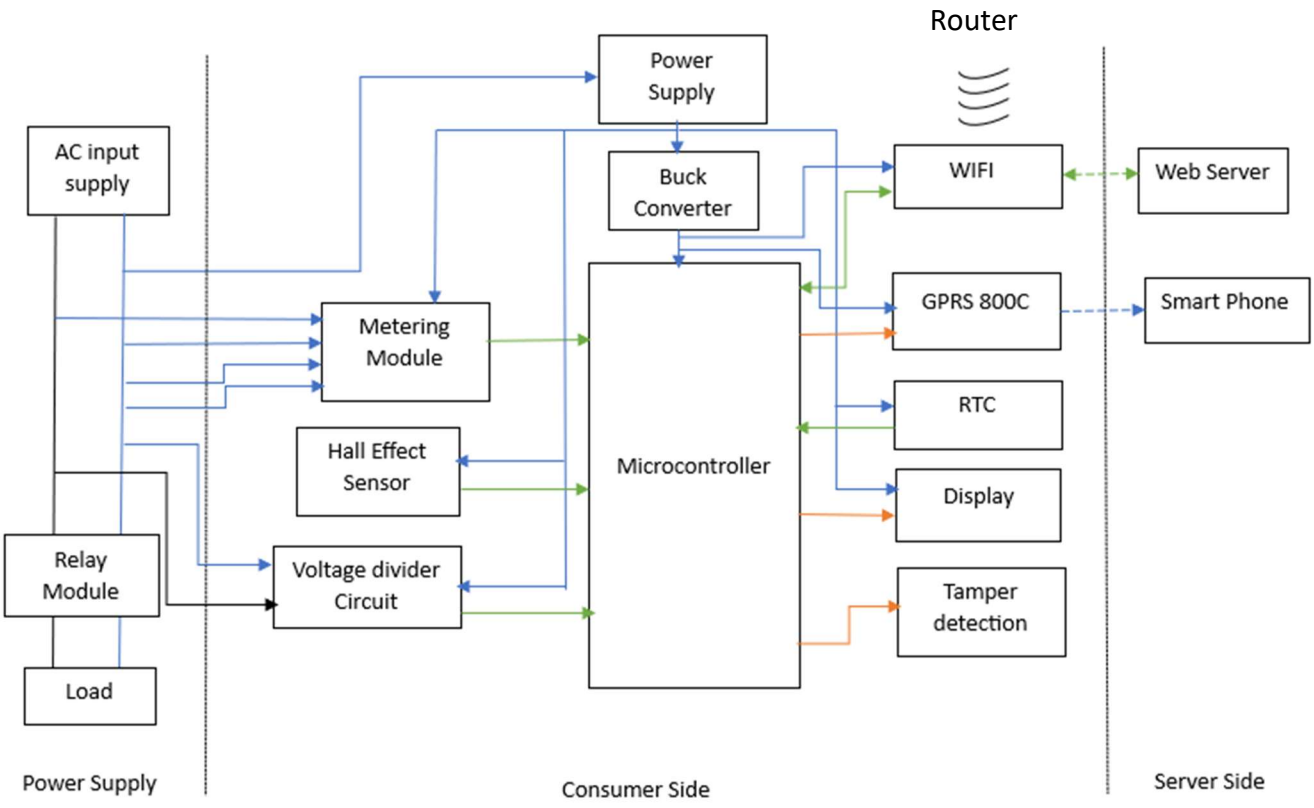


Fig 3.3 Block Diagram: Consumer side

The system begins at the consumer side, where the AC input powers both the load (consumer appliances) and the metering module. The PZEM-004T metering module measures key electrical parameters such as voltage, current, power, and power factor using an internal current transformer (CT) and an external voltage transformer. This data is then sent to the ESP32 microcontroller, which processes and formats it for further transmission. The microcontroller communicates with the utility provider's server via Wi-Fi, enabling real-time monitoring and remote control. In addition, the GPRS module (SIM800L) is used to send alerts such as low balance warnings and tamper notifications to the user. The LCD display provides real-time energy consumption, balance, and other relevant information to the user, ensuring transparency and control. The relay module is used to disconnect the power supply when the user's balance reaches zero, ensuring the prepaid functionality of the system. A tamper detection mechanism monitors for unauthorized access or tampering, and if detected, an alert is sent via the GPRS module.

he Real-Time Clock (RTC) provides precise timestamps for each data entry, ensuring accurate logging and billing The system architecture is designed to be scalable, allowing for future enhancements such as integration with IoT platforms, advanced data analytics, and support for renewable energy sources.

B. Utility side:

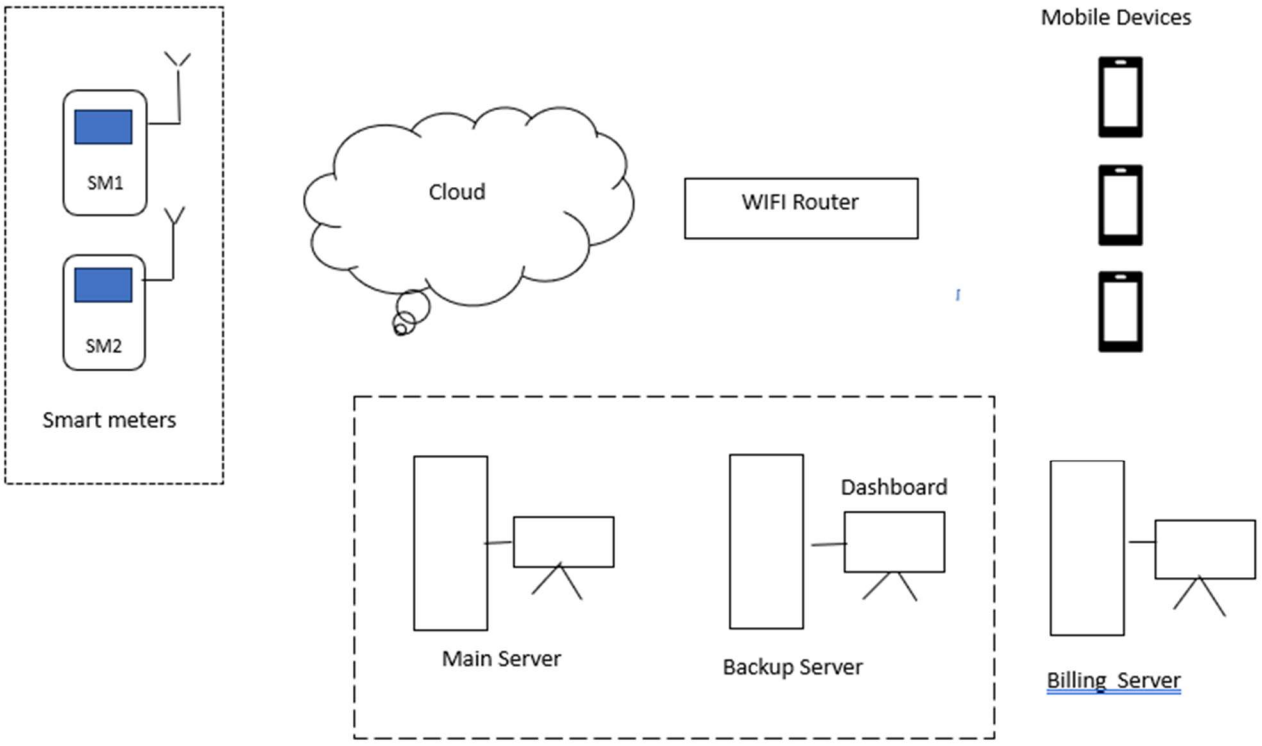


Fig 3.4 Block Diagram: Utility Side

On the utility-side, the GPRS router receives data from multiple smart meters and routes it to the utility's internal network. The database server stores all energy consumption data, billing information, and customer records, providing a centralized repository for data analysis and management. The backup server maintains redundant copies of the data to ensure system reliability and quick recovery in case of failure. The billing server automates billing calculations based on energy usage, updates user accounts, and sends payment reminders. The dashboard interface provides utility providers with real-time insights into energy consumption patterns, allowing them to monitor and manage meters remotely. This interface also enables utility providers to control meters (e.g., disconnect or reconnect) and analyze usage trends for better energy distribution management.

3.7 Specifications of the Proposed System

- Meter Type: Single Phase AC (Residential purpose)
- Operating Mode: Prepaid
- Measurement Range:
  Voltage: 220-250 V
  Current: 0-30A
  Frequency: 50 Hz
  Power Factor Range: 0.95-1
- Display:
  Liquid Crystal display (16x2)
- Communication range (Wi-Fi):
  Using Router: 30-50 meters
  Using mobile Wi-Fi: 8-12 meters.
- Tamper Detection: Magnetic Detection Range:
  Operate point ~175 Gauss
  Release point ~85 Gauss

3.8 Comparative study of the components:

Energy metering module

Parameter	PZEM-004T	HLW-8012
Measure Type	Voltage,Current,Active Power and Energy,Frequency,Power factor	Voltage,Current,Active Power,Energy
Communication	UART	SPI
Voltage Measurement Range	80 to 260V AC	80 to 260V AC
Current Measurement Range	0 to 100A	0 to 10A
Accuracy	0.5%	1%
Power Measurement Range	0 to 22KW	0 to 4.4KW
Operating voltage	5V DC	3.3-5V DC

Table1. Comparison between PZEM-004T and HLW-8012 metering module

From this comparative analysis we concluded that PZEM-004T is ideal for smart energy applications because it supports real-time monitoring of energy consumption data, providing precise and accessible measurements directly. In contrast, the HLW-8012 requires additional processing to convert its PWM signals into usable data, making it less efficient for applications needing immediate and straightforward data integration.

RTC module

Parameter	DS3231	DS1307
Accuracy	±2 ppm (0.432 seconds/day)	±20 ppm (1.728 seconds/day)
Temperature Compensation	Yes, built-in temperature compensation	No temperature compensation
Operating Voltage	2.3V - 5.5V	4.5V - 5.5V
Battery Backup	Supports battery backup with a built-in temperature compensated crystal oscillator	Supports battery backup without temperature compensation
EEPROM/Memory	236 bytes of SRAM	No built-in SRAM
Clock Stability	Very stable across temperature changes	Stability is affected by temperature changes

Table2. Comparison between DS3231 and DS1307

From this comparative analysis we concluded that DS3231 is preferred because it offers higher accuracy (+2 ppm, or 0.432 seconds per day) compared to the DS1307 (+20 ppm, or 1.728 seconds per day). Additionally, it has built-in temperature compensation, ensuring stable timekeeping across temperature changes, which the DS1307 lacks. The DS3231 also includes 236 bytes of SRAM for data storage, providing more functionality and reliability.

Communication Module

Parameter	SIM800C	SIM900A	Quectel M95
Frequency band	Quad band	Dual band	Quad band
Data rate	Up to 85.6 Kbps (DL), 42.8 Kbps (UL)	Up to 85.6 Kbps (DL), 42.8 Kbps (UL)	Up to 85.6 Kbps (DL), 42.8 Kbps (UL)
Supply Voltage	3.4 V to 4.4 V	3.4 V to 4.4 V	3.4 V to 4.4 V
Sleep mode Current	Low (~0.7mA)	Around 1.5mA	Around 1.5mA
Size	Smaller and compact	Large	Large

Table3.Comparision between SIM800C , SIM900A,QuectelM95

The SIM800C module was selected, while all three modules offer similar data rates and operate on the same supply voltage, SIM800C provides quad-band support for better global coverage and has a smaller, more compact form factor, making it ideal for embedded applications. Additionally, it features lower sleep mode current (~0.7mA), enhancing the overall energy efficiency of the system. These advantages make SIM800C a more suitable choice.

Magnetic tamper detection sensor

Parameter	Hall effect sensor	Reed switch sensor
Working Principle	Detects magnetic fields through changes in voltage across a semiconductor.	Mechanical switch that opens/closes in response to a magnetic field.
Sensitivity to Magnetic Field	High sensitivity to weak magnetic fields.	Lower sensitivity, requiring a stronger magnetic field.
Response Time	Fast, typically in the microsecond range.	Slower response time due to mechanical movement.
Tamper Detection Accuracy	High precision for detecting small magnetic tampering attempts.	Less precise, can miss weak or subtle tampering.
Reliability	Highly reliable, especially in harsh environments.	Mechanical wear can reduce reliability over time.

Table4. Comparison between Hall effect sensor and Reed Switch

From this comparative analysis we concluded that the Hall effect sensor due to its high sensitivity to weak magnetic fields, which allows it to detect even subtle magnetic tampering attempts with high precision. It has a fast response time, typically in the microsecond range, which enables quick detection and improves the reliability of the anti-tampering system. Although it has higher power consumption than a reed switch, its accuracy and sensitivity make it a better choice for applications where precise and reliable tamper detection is essential.

Relay Module:

Parameter	SK90-S-DC5V-1C	SRD-05VDC-SL-C
Contact Rating	10A 250VAC / 10A 30VDC	10A 250VAC / 10A 30VDC
Response Time	Faster switching (5–10ms)	Slower (~10ms)
Reliability	Higher mechanical durability (better for repeated operations)	Average
Size	More compact	Slightly bulkier
Optoisolation	Yes (reduces interference to microcontroller)	Often not present in cheaper versions

Table5. Comparison between SK90-S-DC5V-1C, SRD-05VDC-SL-C

The SK90-S-DC5V-1C relay module was selected for this project due to its faster switching response and compact form factor, which helps in saving space on the circuit board. Additionally, it features optoisolated input, providing better electrical isolation and minimizing noise interference in sensitive control circuits. Its compatibility with both 3.3V and 5V logic makes it suitable for seamless interfacing with modern microcontrollers like Arduino and ESP modules.

Power Supply

Parameter	Hi-Link HLK-5M05	Generic 5V AC-DC Adapter	HLK-PM01
Input and Output Voltage	90–264V AC, 5V DC	100–240V AC, 5V DC	90–264V AC, 3V DC
Efficiency	~85%	~70–80%	~75–80%
Size	Very Compact (33x19x15mm	Bulky (external adapter type)	Compact (37x23x18mm)
Heat Dissipation	Low	Medium	Medium
Electrical Isolation	Yes	Yes	Yes

Table6. Comparison between Hi-Link HLK-5M05,Generic 5V AC-DC Adapter,HLK-PM01

The Hi-Link HLK-5M05 was selected because it offers higher efficiency (~85%), higher power output (5W), compact PCB-mountable design, and international safety certifications. These features make it highly suitable for embedded applications like smart energy meters, ensuring both reliability and safety in direct AC-powered environments.

3.9 Specifications of the selected components:

1. PZEM-004T METERING MODULE: -



Fig 3.5 PZEM-004T metering module

Specifications:

- Voltage Range: 80–260V,
- Current Range: 0–10A or 0–100A, Starting Current: 0.01A (10A), 0.02A (100A)
- Power Factor Range: 0.00–1.00, Accuracy: 1%.
- Power Range: 0–2.3kW (10A) or 0–23kW (100A), Resolution: 0.1W.
- Operating Temperature: -20°C to +60°C
- Power Requirement: Approximately 0.5W

2.DS3231 RTC:



Fig 3.6 DS 3231 RTC module

- Accuracy: ±2 minutes/year (0°C–40°C), ±3.5 minutes/year (-40°C–85°C).
- Battery Backup: CR2032, keeps time during power outages.
- Interface: I2C, Speed: Up to 400kHz.
- Operating Voltage: 3.3V–5.5V
- Power Requirement (worst case): P = 5 V × 0.002 A = 0.01 W

3.ESP 32 WROOM 32 MCU:



Fig 3.7 Esp WROOM32

- Processor: Dual-core Xtensa LX6, Clock Speed: 240 MHz
- Memory: 520KB SRAM, up to 8MB Flash.
- Connectivity: Wi-Fi (802.11 b/g/n) and Bluetooth (BLE).
- GPIO Pins: 34 GPIO, ADC: 12-bit SAR ADC with up to 18 channels.
- Operating Temperature: -40°C to 125°C
- Power Requirement: P = V × I = 3.3 V × 0.16 A = 0.528W

4.A1220 Hall effect sensor:



Fig 3.8 A1220 Hall Effect sensor

• Operating Voltage: 3.3V–24V.
• Response Time: 2µs, Supply Current: 3.5mA at 5V.
• Magnetic Sensitivity: ±65 Gauss, Output Type: Digital (open-drain)
• Power Requirement: P = 5 V × 0.015 A = 0.075 W

5.GPRS 800C



Fig 3.9 GPRS 800C

• Frequency Bands GSM850, EGSM900, DCS1800, PCS1900
• Data Rate Up to 85.6 kbps (download), 42.8 kbps (upload)
• Supply Voltage 3.4 V to 4.4 V
• Current Consumption
    Power Down 60 uA
    Sleep Mode 1 mA
    Standby 18 mA
    GPRS transmission 453 mA
    Transmission Burst Up to 2A
• Operating Temperature -40°C to +85°C

6. LCD display:

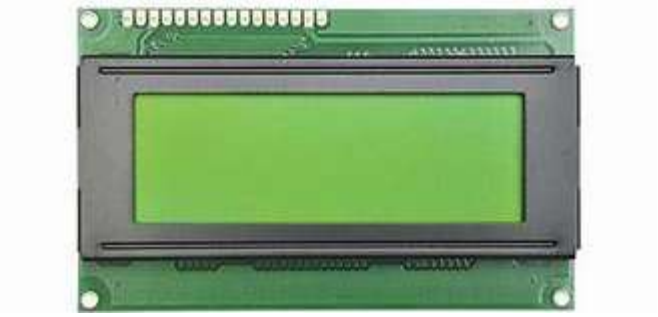


Fig 3.10 LCD display

• Display Size: 20 characters per line, 4 lines
• Character Size: Typically, 5x8 pixel matrix per character
• Operating Voltage: 4.7V to 5.3V (standard 5V)
• Current Consumption: Approx. 1-2 mA (without backlight), up to 25 mA with backlight

7. Relay module (SK90-S-DC5V-1C)



Fig 3.11 Relay Module

- Coil Voltage: 5V DC.
- Contact Rating: 10A at 250V AC / 30V DC.
- Switching Power: 2500VA (AC), 300W (DC).
- Switching Time: Operate: 10ms, Release: 5ms.
- Operating Temperature: -40°C to +85°C.
- Power Requirement: P = 5 V × 0.015 A = 0.075

8.Hi-Link HLK-5M05 5V SMPS



Fig 3.12 Power Supply

- Input Voltage: 90–264V AC or 100–370V DC.
- Output Voltage: 5V DC, Current: 4A (20W).
- Efficiency: Up to 80%, Ripple & Noise: ≤50mV.
- Protection: Short-circuit, over-current, over-voltage.
- Isolation Voltage: 3000V AC, Certifications: CE, UL, RoHS.
- Power Requirement: P = 5 V × 4 A = 20 W

3.10 Software Design:

- Database Schema: Models: Users, Consumption, Billings, Alerts, Admins.
- Workflow:
    a) User logs in and accesses the user dashboard.
    b) Admin logs in and accesses the admin dashboard.

c) Data is fetched from the backend and displayed dynamically on both dashboards.

A. User Dashboard Implementation:

- Navbar: Contains links to Dashboard, Billing, Alerts, Current-Data, Profile, Support, and Logout.
- Dashboard:
- Displays real-time consumption data.
- Provides options for recharging.
- Shows daily, weekly, and monthly usage in graphical format.
- Billing: Displays billing history with details like date, amount, and payment status.
- Alerts: Shows notifications for low balance or unusual usage patterns.
- Current Data: Displays live electricity consumption data.
- Profile: Allows users to update their personal information after verification.
- Support: Provides a contact for user assistance.
- Logout: Securely logs the user out of the system.

B. Admin Dashboard Implementation

- Navbar: Contains links to Dashboard, Manage Users, Billing History, Alerts, Analytics, Profile, Support, and Logout.
- Dashboard: Displays key metrics: Total Users, Active Energy Meters, and Alerts Pending.
- Manage Users: Shows a table with User ID, MAC ID, Name, Last Update, Total Consumption (kWh), Remaining Balance, Relay Status (ON/OFF), and Action (ON/OFF) for all users.
- Billing History: Displays billing history for all users.
- Alerts: Shows alerts related to low balance and tampering.
- Analytics: Displays Peak Consumption Hours and Average Monthly Consumption.
- Profile: Allows admins to update their personal information.
- Support: Provides a for admin assistance.
- Logout: Securely logs the admin out of the system.

3.11 Calculations:

A. Power Calculations:

Total Power=PPZEM+PESP32+PDS3231+PAT24C256+PLCD+PSIM800L+PZMPT101B +PA3144+PD2440+PTCA9548A

Substituting the power requirement values:

Total Power=0.5+0.53+0.01+0.005+0.1+8+0.1+0.075+0.075+0.05=9.445W

B. Storage calculations:

Payload Fields
:mac_id(string)+timestamp(date)+consumption(float)+relay_status(string)+voltage(float)+current(float)+power(float)+balance(float)+power_factor(float)+frequency(float)+reset_time(date)+update_time(date).
Total payload size :
12+8+8+5+8+8+8+8+8+8+8+8=93 bytes
The SPIFFS (SPI Flash File System) of ESP32 is 1MB (1,048,576 bytes), and each entry store takes 93 bytes.
To find the total number of entries to store :
1,048,576 bytes/93 bytes=11,277.15 Total Time = 11,277 entries × 10seconds/entry = 112,770seconds
Now, converting seconds to hours: 112,770 seconds/(3600seconds/hour)=31.3 hours
So entries stores for approximately 31 hours and 18 minutes .

Alert payload Fields :
mac_id(string)+user_id(string)+alert(string)+time_stamp(date).
Total size per record: 12 + 12 + 50 + 8 = 82 bytes
total data size for one entry 93 bytes + 82 bytes = 175 bytes.
Let's calculate how many entries can be stored with the total 1MB (1,048,576 bytes) of SPIFFS storage:
Total entries= Total storage/ Total storage = 1,048,576bytes / 175 bytes/entry = 5,997 entries
Now, for the time duration:
Since each entry takes 10 seconds, the total time for 5,997 entries is:
Total Time= 5,997entries×10seconds/entry = 59,970seconds
Converting to hours: 59,970seconds / 3600 seconds /hour = 16.66hours
So entries stores for approximately 16 hours and 40 minutes .

For storing balance in NVS:
Balance +Consumption = 80 bytes total

3.12 PCB Schematic

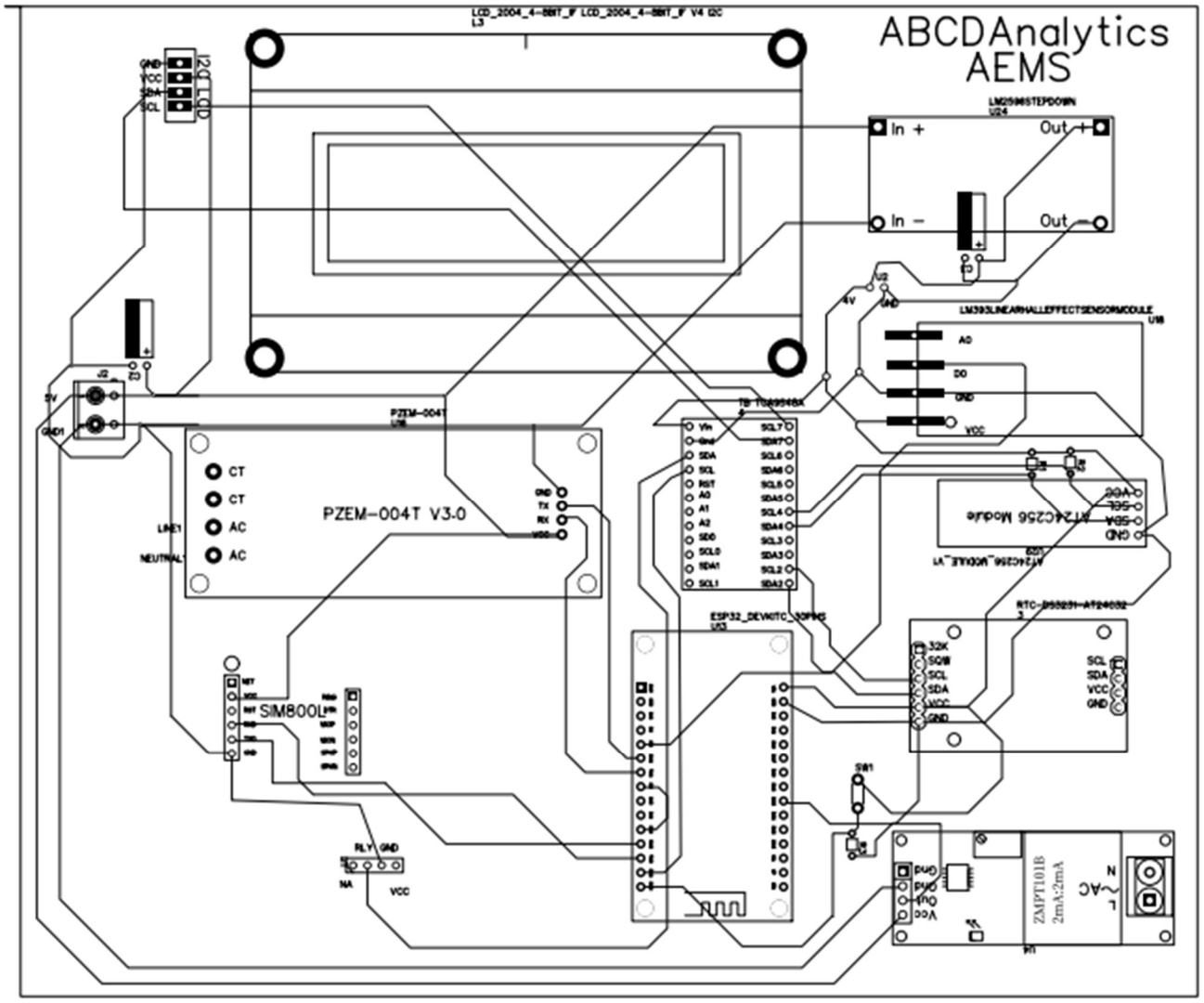


Fig3.13 PCB Schematic

Chapter 4
SIMULATION AND TESTING

4.1 Simulation result:

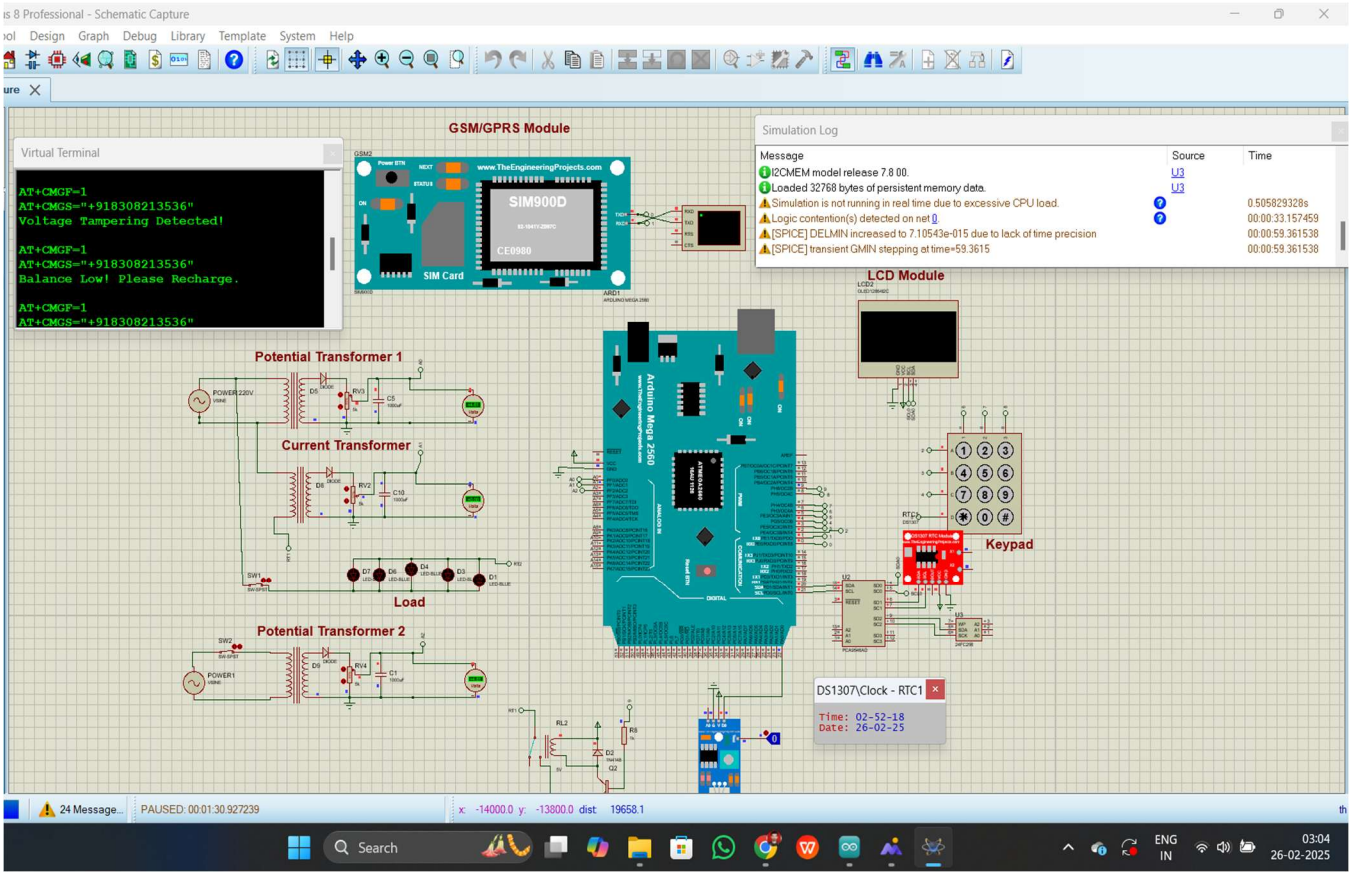


Fig4.1 Proteus Simulation

Fig 4.1 The simulation in Proteus software demonstrates the core functionalities of a Smart Energy Meter using an Arduino Mega as a substitute for ESP32, which is not supported in Proteus. The system monitors energy consumption by measuring voltage and current using Potential Transformers (PTs) and Current Transformers (CTs) instead of a PZEM sensor, which is unavailable in Proteus. The measured values are used to calculate power and track energy consumption over time. A GPRS module (SIM900D) is included to enable communication, sending SMS alerts for low balance warnings, power disconnection, and voltage tampering detection. If tampering is detected using a voltage sensor, the system immediately notifies the user via SMS. The balance and billing system deducts a predefined cost per unit (1 unit = 200 watts) and automatically disconnects the relay when the balance reaches zero. A 20x4 LCD module provides real-time updates on voltage, current, energy usage, and balance, while a Real-Time Clock (RTC - DS1307) maintains an accurate record of date and time for tracking energy consumption and tamper events. Since the ESP32-based server communication for balance recharge is not feasible in this simulation, a 4x4 keypad allows manual balance entry, simulating the recharge process. This simulation effectively validates the prepaid smart energy meter concept, demonstrating key features such as energy monitoring, automated

billing, remote alerts, and tamper detection, which will be further enhanced in the actual ESP32-based implementation.

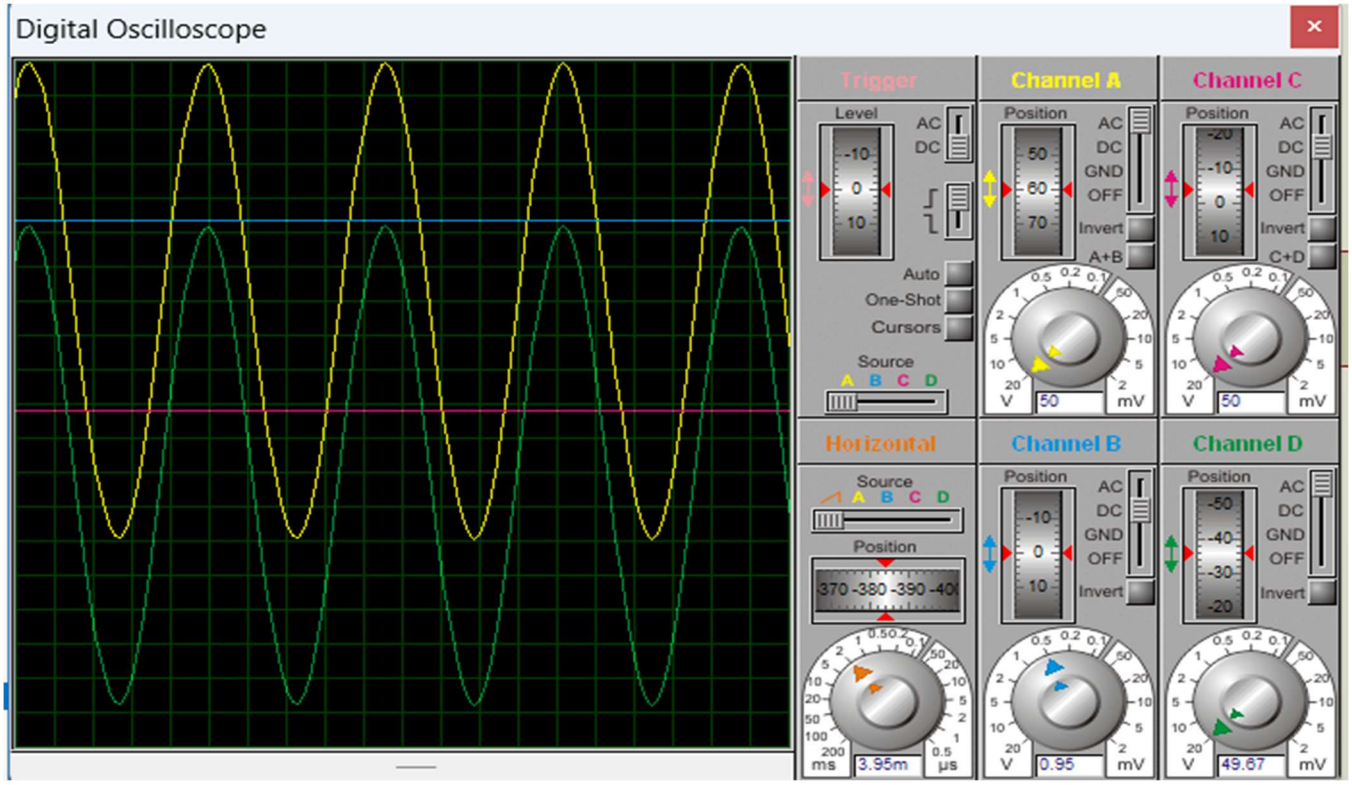


Fig 4.2 Simulation Waveforms

- The waveforms for the PT and CT outputs closely follow the shape of the AC mains waveform, indicating accurate signal transformation and measurement.
- The PT output (pink) is in phase with the AC reference (green), confirming proper voltage transformation.
- The CT output (blue) is also sinusoidal and in phase, demonstrating correct current sensing and minimal phase shift, which is essential for precise power and energy calculations.

4.1.1 Simulation output data:

Sr.no	Load (ohm)	Input Voltage (V)	Current (A) CT	Voltage(V) PT	Calculated Current (I=V/R)
1	400	230	0.580	5.03	0.570
2	450	230	0.523	5.02	0.511
3	500	230	0.470	5.04	0.460
4	1000	230	0.235	5.05	0.230

Table 7. Readings based on simulation

4.1.2 Calculations:

A. Current Calculations

$$\text{Current Deviation (\%)} = \frac{\text{Measured (CT)- Calculated}}{\text{Calculated}} \times 100\%$$

1. 400 Ω:
   $= [(0.580-0.570)/0.570] \times 100$
   $=1.75\%$
2. 450 Ω:
   $= [(0.523-0.511)/0.523] \times 100$
   $=2.35\%$
3. 500 Ω:
   $= [(0.470-0.460)/0.460] \times 100$
   $=2.17\%$
4. 400 Ω:
   $= [(0.235-0.230)/0.230] \times 100$
   $=2.17\%$

B. PT Ratio Calculation:

$$\text{PT Ratio} = \frac{\text{Input Voltage}}{\text{PT Output}}$$

1. 400 Ω: $230/5.03 \approx 45.7$
2. 450 Ω: $230/5.02 \approx 45.8$
3. 500 Ω: $230/5.04 \approx 45.62$
4. 1000 Ω: $230/5.05 \approx 45.5$

The PT ratio remains consistent around 45.5–45.8, indicating the PT is functioning as expected.

The simulation results demonstrate that the measured current values using the current transformer (CT) are in close agreement with the theoretical values calculated using Ohm's Law, with deviations ranging from 1.75% to 2.35%. This level of accuracy is within or very close to the permissible limits for Class 2 metering applications, indicating reliable current measurement. The potential transformer (PT) output consistently reflects a step-down ratio of approximately 1:45.7, confirming accurate voltage transformation and measurement.

4.2 Hardware Testing

Testing Methodology:

- Testing a hardware system using a Digital Multimeter (DMM)
- Connection verification.
- Individual Component Testing
- Voltage across Each Component

- Current requirement of each component
- Check for overheating.

4.2.1 Component-wise Voltage and Current Measurements

Sr. No.	Component	Input	Output
1	Power Supply	Input voltage : 245.1 V	Output voltage: 5.04 V
2	Relay	Input voltage : OFF: 3.001 V ON: 10.4 mV	
3	PZEM	Input voltage : 4.93 V	Input AC Voltage: 245.1 V
4	Buck converter	Input voltage : 4.88 V	Output voltage: 4.04 V
5	ESP32	Input voltage : 4.02 V Current:140 mA	
6	Hall sensor	Input voltage : 4.02 V Current :10 mA	Output voltage: without magnet : 3.45 V with magnet: 017.4 mV
7	RTC	Input voltage : 4.02 V Current:1.7 mA	Output voltage: 4.02 V
8	TCA mux	Input voltage : 4.02 V Current : 10 mA	Output voltage: 4.9 V
9	ZMPT voltage sensor	Input voltage : 4.97 V Current : 15 mA	Output voltage: 1.20 V
10	SIM 800L	Input voltage : 4.95 V Current : 18 mA	Output voltage: 3.103 V
11	LCD	Input voltage : 4.98 V Current : 12 mA	Output voltage: 2.45 V

Table8.Component-wise Voltage and Current Measurements

Power Supply Regulation Testing :
The Hi-Link SMPS module used in this project was tested for both line and load regulation. The output voltage remained stable within 0.4% across the input voltage range of 100V to 260V AC, and within 0.2% as the load current varied from no load to full load. These results confirm the suitability of the Hi-Link SMPS for reliable operation of the project.

Calculated Line Regulation:

Line Regulation (%) = ((Vout, max−Vout, min)/Vnominal) ×100
= (5.01-4.99/5) ×100
=0.4%

Load Regulation (%)=((Vno-load−Vfull-load)/ Vnominal)×100
= (5.00-4.99)/5))×100
=0.2%

4.2.2 System Performance testing:

Load ratings	Readings By PZEM	Measured by Tong tester & DMM
Cooler (190 W)	Voltage :235.6 V Current :0.62 A Power : 0.15 Kw	Voltage :236.6 V Current :0.61 A Power : 0.145 Kw
Bulb (18 W)	Voltage :235.9 V Current :0.08 A Power : 0.020Kw	Voltage :237.6 V Current :0.1A Power : 0. 023Kw
Bulb(200W)	Voltage :234.30V Current :0.67 A Power : 0.16 Kw	Voltage :237.6 V Current :0.1 A Power : 0. 17Kw
Bulb (200+18)W	Voltage :235.30V Current :0.75 A Power : 0.170 Kw	Voltage :236.30V Current :0.70 A Power : 0.165 Kw
2 Bulb +Cooler (18+200+190)W	Voltage :236.30V Current :1.36 A Power : 0.321 Kw	Voltage :237.0V Current :1.40 A Power : 0.331 Kw

Table 9. System performance testing

System Accuracy calculations:

$$\text{Deviation (\%)} = \frac{\text{PZEM Reading} - \text{Measured Reading}}{\text{Measured Reading}} \times 100\%$$

For Bulb (200 W)

Deviation (%) = [(0.160−0.170) /0.170] ×100%

= -5.5%

For 408W

Deviation (%)= [(0.321−0.331) /0.331] ×100%

= 3.02%

For 218W

Deviation (%)= [(0.170−0.165) /0.165 ×100%

= 3.03%

Calculated deviations for the Bulb (200 W), 408 W, and 218 W loads are -5.5%, 3.02%, and 3.03% respectively. According to IS 13779 and MSCB/MSEDCL standards for residential single-phase energy meters, the permissible error is typically within ±2% for Class 2.0 meters and ±1% for Class 1.0 meter

The observed deviations for the tested loads are slightly above the permissible limits specified by IS 13779 for residential energy meters. This can be attributed in part to the inherent limitations of the tong tester, which does not provide high-resolution current measurements at lower load conditions. As a result, while the PZEM module offers reasonable accuracy for general monitoring, improvements in both meter calibration and reference instrumentation would be necessary to achieve compliance with residential metering standards and ensure reliable, certified billing performance.

4.3 Software Testing:

Testing Methodology:

- Unit testing for individual components (e.g., recharge functionality, relay control).
- Integration testing for data flow between frontend and backend.
- User acceptance testing (UAT) for both user and admin dashboards.
- Tools Used: Hopscotch: Used for API testing.
- Test Cases:
    a) Verify that the admin can toggle the relay status for a user.
    b) Verify that the User can recharge.
    c) Verify consumption graphs.

4.3.1 Dashboard interfaces

A. Admin Dashboard:

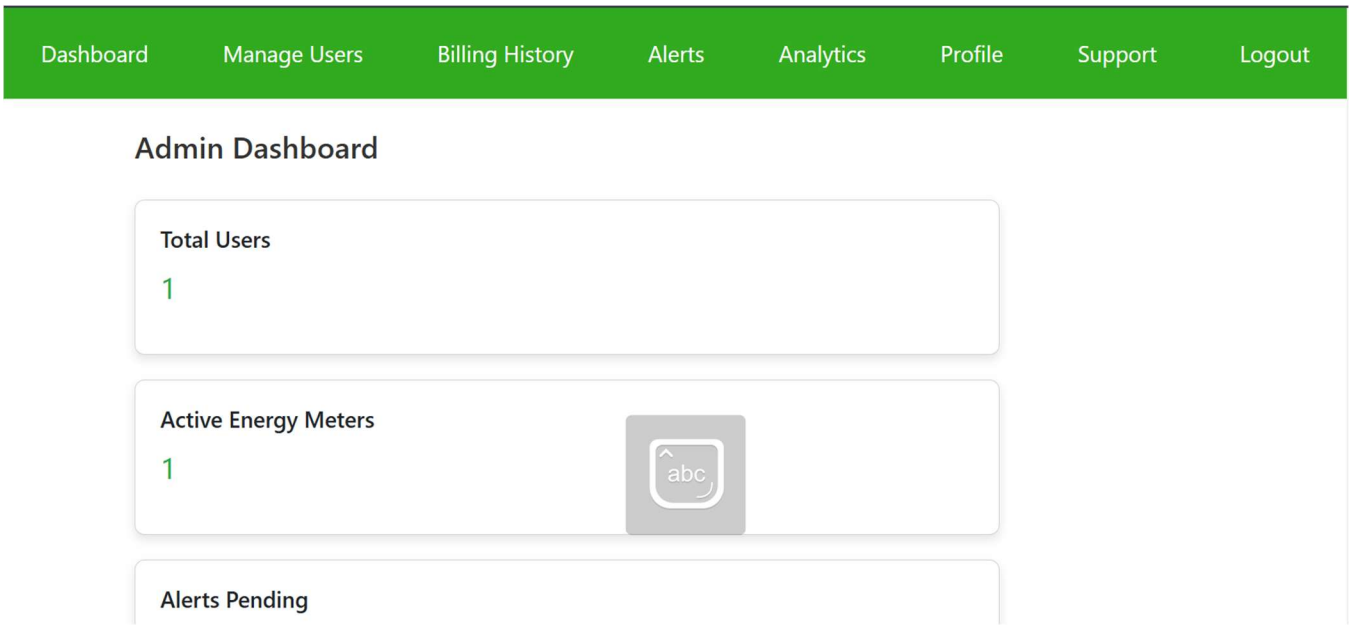


Fig 4.3 Admin Dashboard

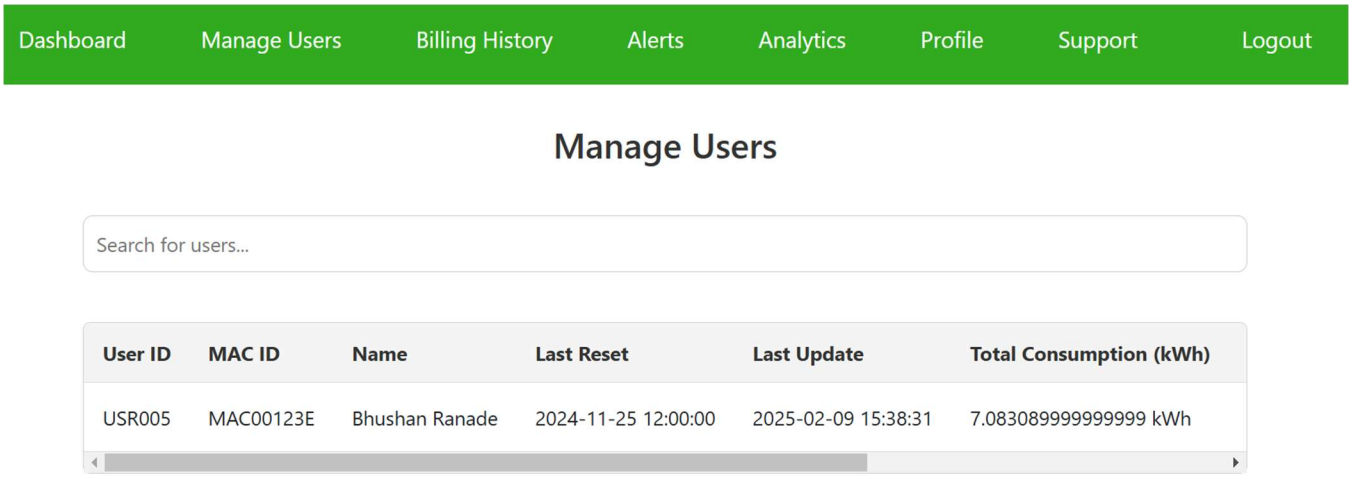


Fig 4.4 Admin Dashboard: Manage users

B. User dashboard:

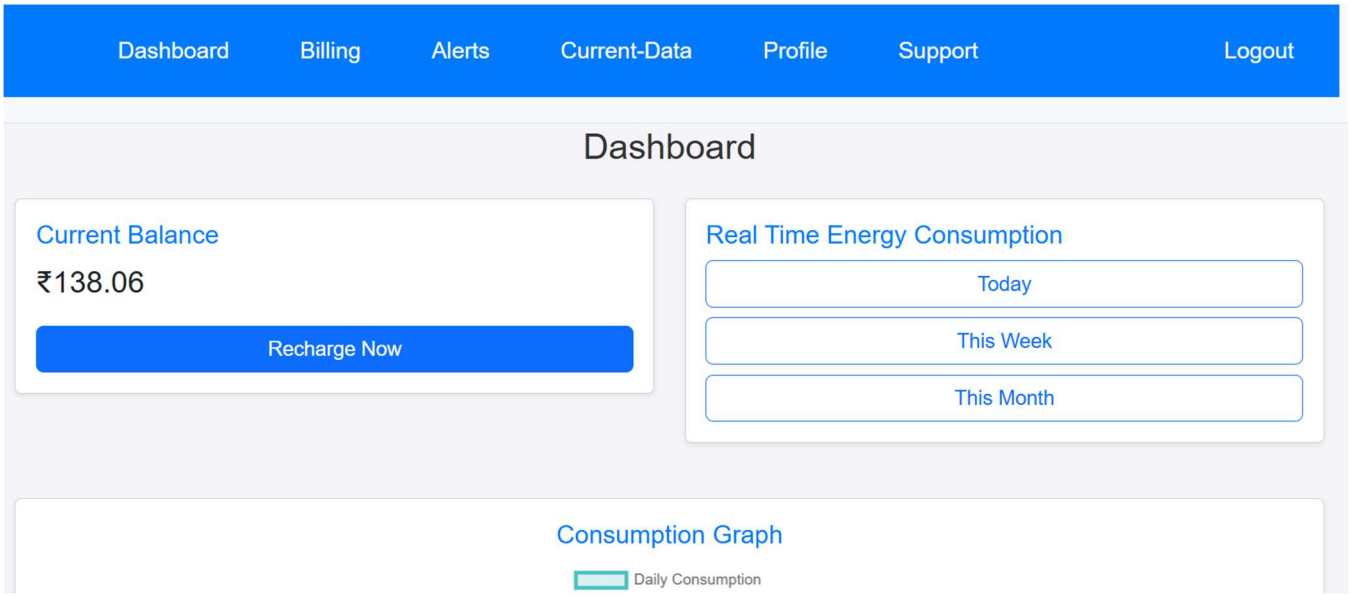


Fig 4.5 User Dashboard

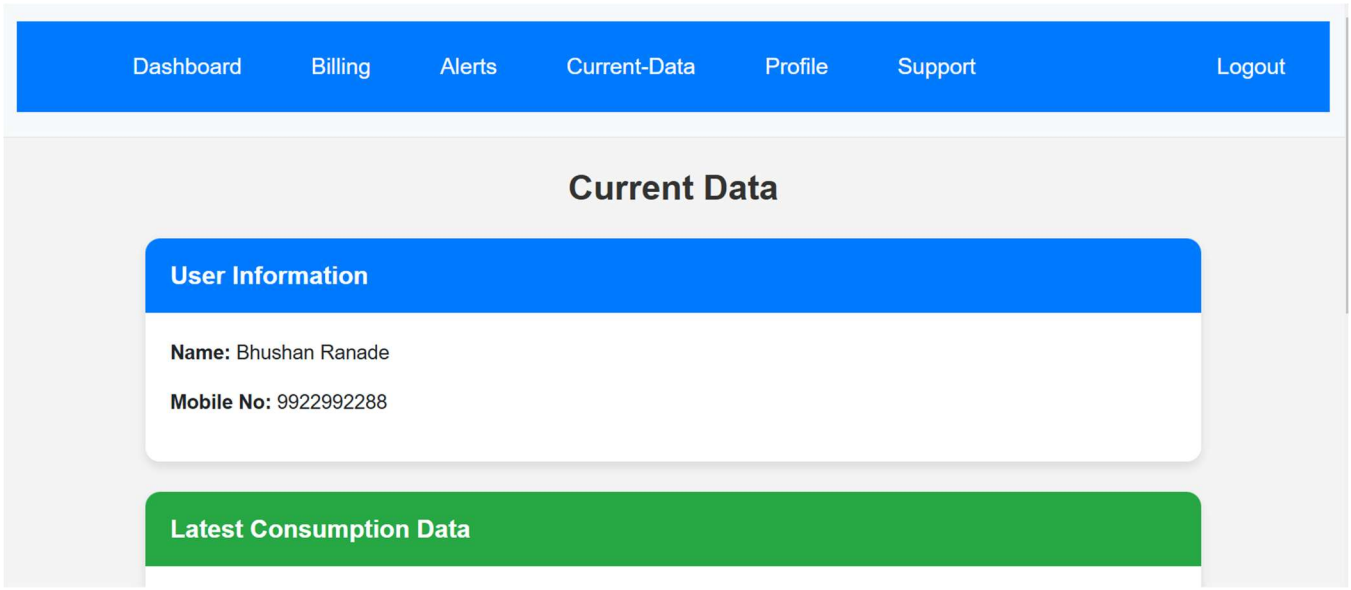


Fig 4.6 User Dashboard: Current data

4.3.2 Meter data logged in database

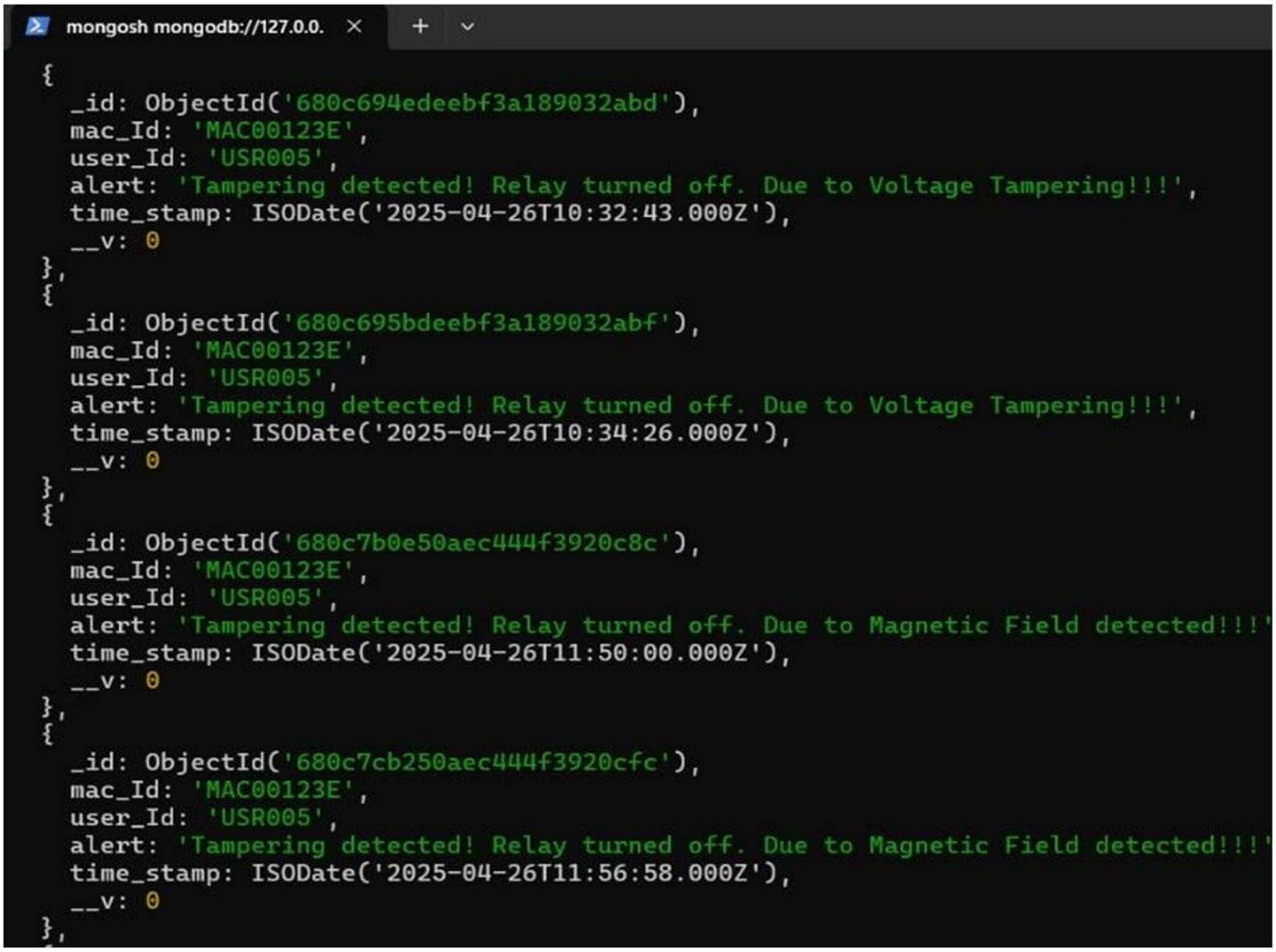


Fig4.7 User registered in database

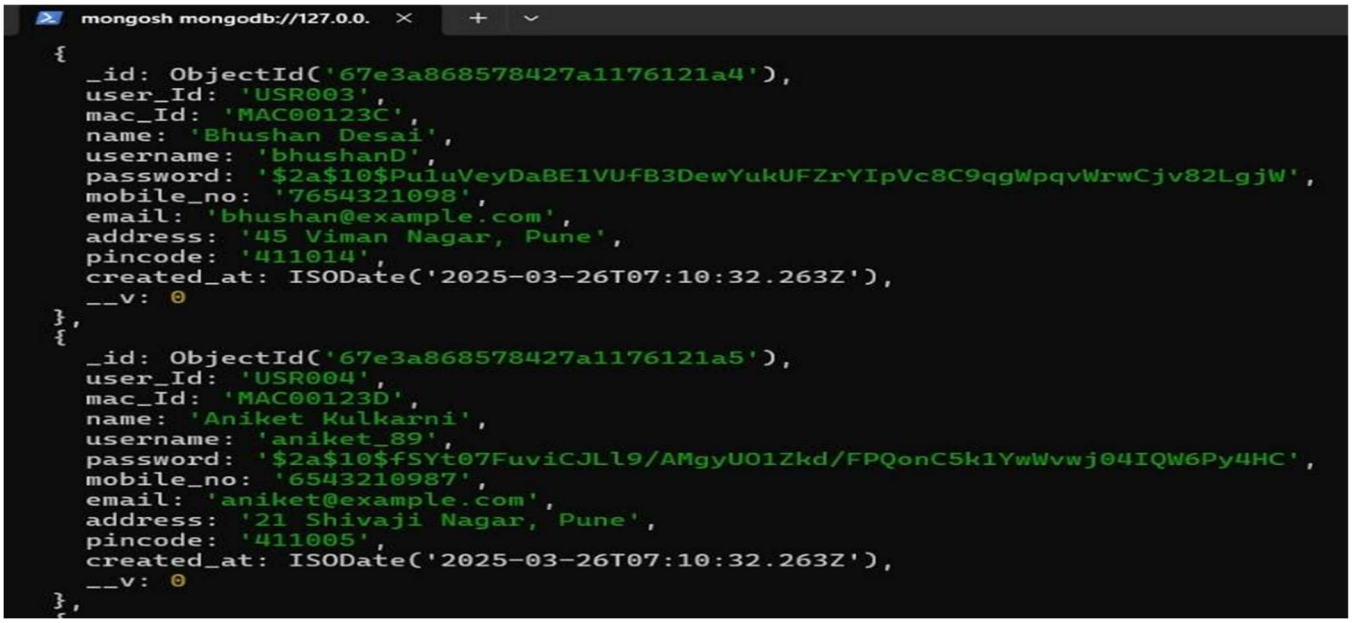


Fig4.7 Stored meter readings in database



Fig4.8 Records of Meter Readings

Chapter 5
RESULT ANALYSIS AND CONCLUSION

5.1 Hardware Setup:

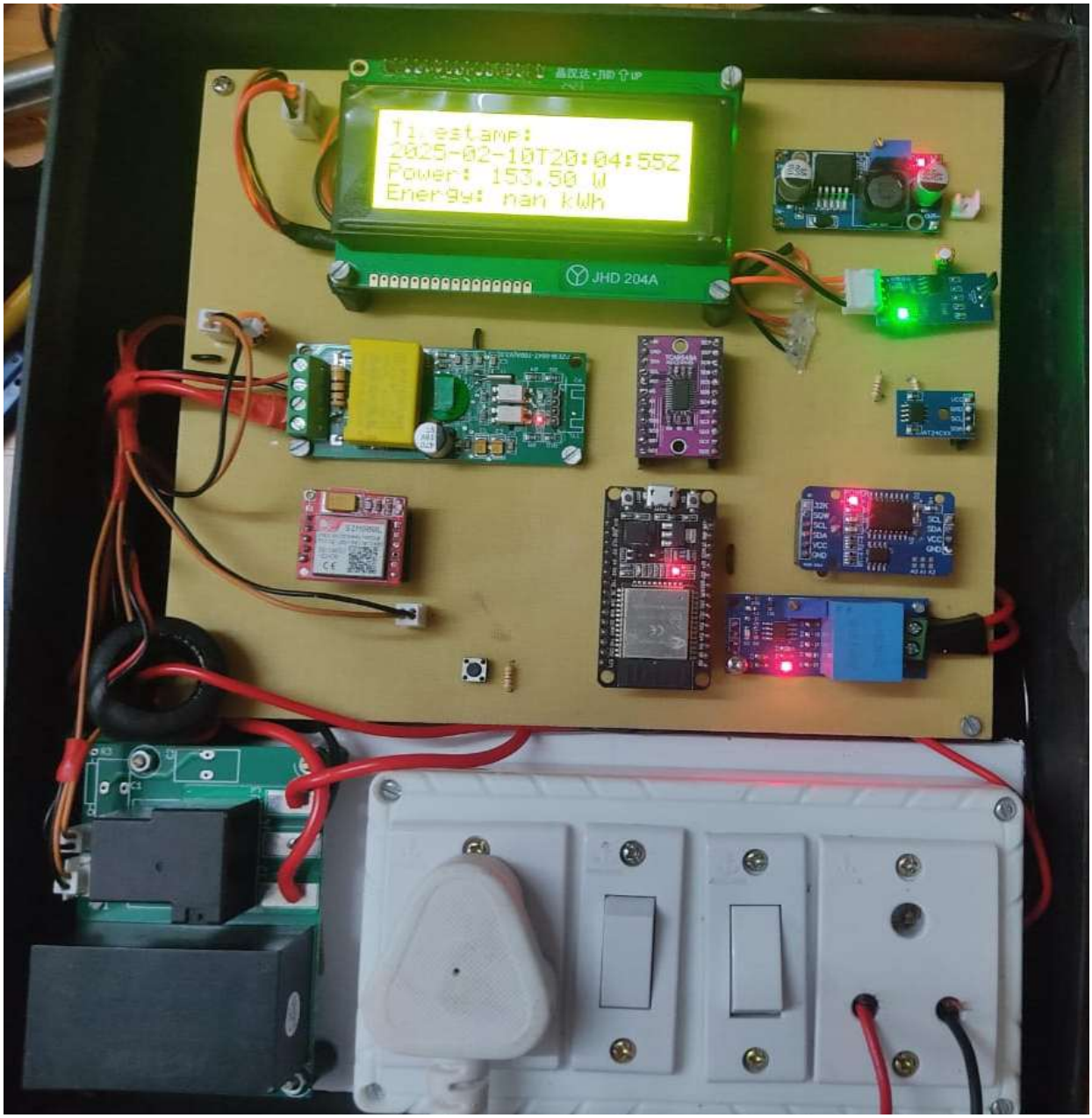


Fig 5.1 Hardware Setup

The comprehensive evaluation of the developed prepaid energy metering system, through both hardware testing and simulation, demonstrates that the system performs reliably and meets the essential accuracy requirements for residential energy metering. Hardware tests comparing the meter readings with reference instruments such as tong testers and digital multimeters showed measurement deviations generally within ±3%, with some variation at very low loads due to the limited resolution of the reference instruments. These deviations, while slightly above the strictest Class 1 accuracy limits, are close to or within the Class 2 (±2%) limits prescribed by IS 13779 and similar metering standards.

Simulation results conducted in Proteus further validated the system's performance under controlled conditions. The current transformer (CT) and potential transformer (PT) outputs closely matched theoretical calculations, with current measurement deviations ranging from 1.75% to 2.35% across various load resistances, and a stable PT voltage ratio around 1:45.7. Oscilloscope waveform analysis confirmed that the PT and CT outputs are sinusoidal and phase-aligned with the AC mains voltage, ensuring accurate power and energy measurement.

The slight discrepancies observed in both hardware and simulation measurements can be attributed to factors such as CT ratio calibration, instrument resolution limitations, and minor circuit losses. Nonetheless, the system's overall accuracy and stability demonstrate its suitability for practical deployment in residential energy monitoring applications. The integration of accurate sensing, real-time data processing, user interface features, and communication modules collectively ensures reliable energy measurement and user-friendly operation.

5.2 Conclusion

The Automated Electricity Metering System with a Prepaid Model offers a smart and efficient solution to modern energy management challenges. With features like real-time monitoring, prepaid billing, mobile payments, and tamper detection, it ensures accurate billing, enhances transparency, and gives users better control over their electricity usage.

The system reduces manual effort, improves security, and supports efficient load management. Future upgrades like IoT integration, AI-based analytics, and cloud dashboards can make the system even smarter and more sustainable. Overall, this project provides a foundation for a reliable and future-ready energy infrastructure.

5.3 Future Scope:

The prepaid energy metering system developed in this project can be further enhanced by integrating smart grid compatibility, mobile app support, and advanced data analytics for improved user experience and energy management. Future developments may include renewable energy monitoring, multi-tariff billing, enhanced security features, and remote firmware updates. These upgrades will make the system more versatile, scalable, and suitable for both residential and commercial applications, keeping pace with the growing demands of smart energy solutions.

Chapter 6
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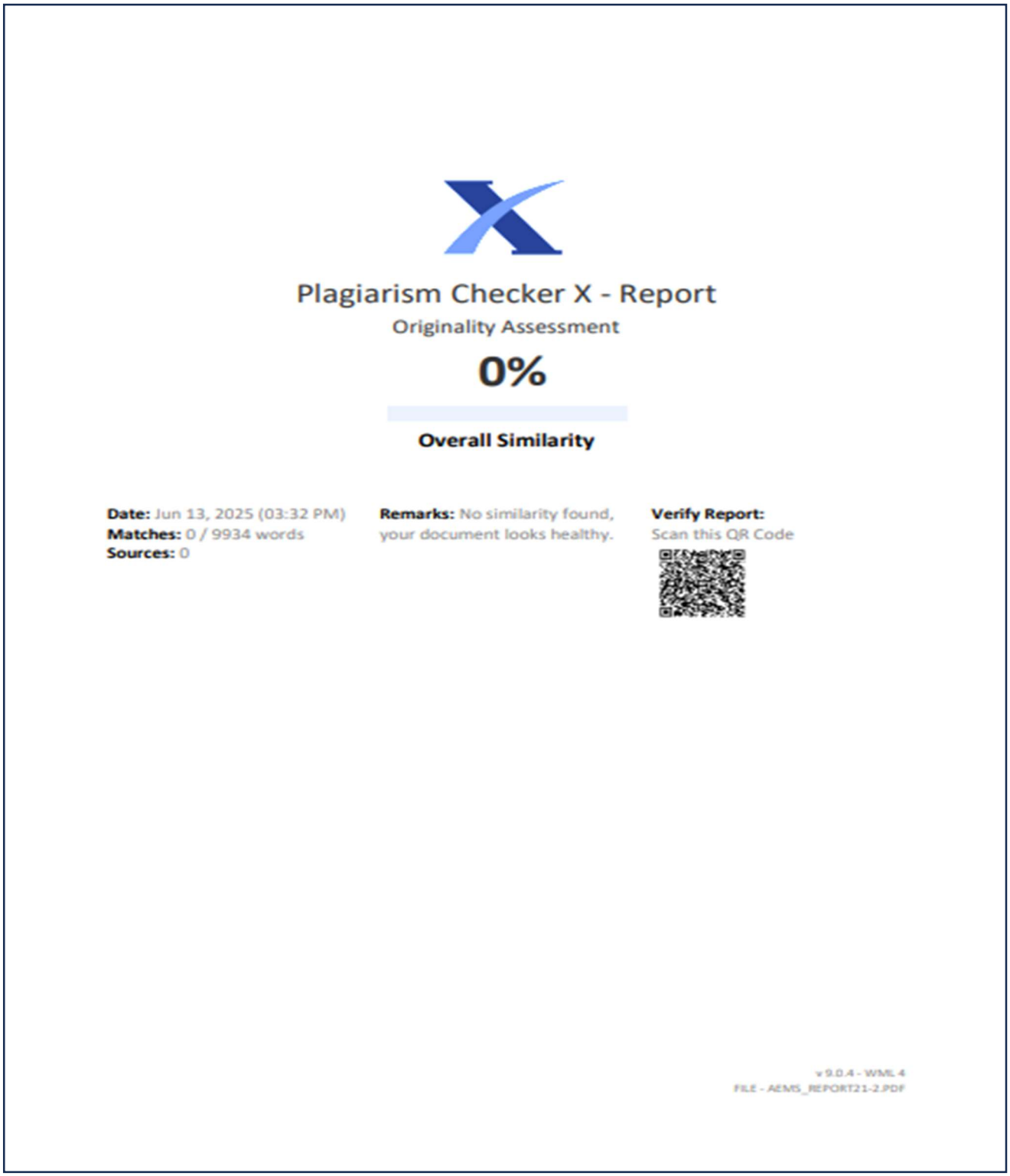
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Bill of Materials:

Module Name	Module No.	Price	Quantity	Total Price	Link
METERING MODULE	PZEM-004T	₹700.00	1	₹700.00	Link
MCU	ESP 32 WROOM 32	₹350.00	1	₹350.00	Link
RTC MODULE	DS3231	₹225.00	1	₹225.00	Link
EEPROM MODULE	AT24C256	₹90.00	1	₹90.00	Link
*4 LCD	JHD629204A	₹450.00	1	₹450.00	Link
GPRS	SIM 800C	₹250.00	1	₹250.00	Link
VOLTAGE SENSOR MODULE	ZMPT101B	₹125.00	1	₹125.00	Link
HALL EFFECT SENSOR MODULE	A1220	₹109.30	1	₹109.30	Link
I2C MULTIPLEXER	TCA9548A	₹200.00	1	₹200.00	Link
PUSH BUTTON	PBS1	₹2.50	1	₹2.50	Link
RELLAY MODULE	SK90-S-DC5V-1C	₹55.00	1	₹55.00	Link
AC to DC Power Supply Module	HLK-20M05	₹350.00	1	₹350.00	Link
Total			13	Rs.3217.20	

Datasheets:
PZEM-004T MODULE

PZEM-004T V3.0 User Manual

Overview

This document describes the specification of the PZEM-004T AC communication module, the module is mainly used for measuring AC voltage, current, active power, frequency, power factor and active energy, the module is without display function, the data is read through the TTL interface.

PZEM-004T-10A: Measuring Range 10A (Built-in Shunt)

PZEM-004T-100A: Measuring Range 100A (external transformer)

1. Function description

1.1 Voltage

1.1.1 Measuring range:80～260V

1.1.2 Resolution: 0.1V

1.1.3 Measurement accuracy: 0.5%

1.2 Current

1.2.1 Measuring range: 0～10A (PZEM-004T-10A)； 0～100A (PZEM-004T-100A)

1.2.2 Starting measure current: 0.01A (PZEM-004T-10A)； 0. 02A (PZEM-004T-100A)

1.2.3 Resolution: 0.001A

1.2.4 Measurement accuracy: 0.5%

1.3 Active power

1.3.1 Measuring range: 0～2.3kW (PZEM-004T-10A)； 0～23kW (PZEM-004T-100A)

1.3.2 Starting measure power: 0.4W

1.3.3 Resolution: 0.1W

1.3.4 Display format:

＜1000W, it display one decimal, such as: 999.9W

≥1000W, it display only integer, such as: 1000W

1.3.5 Measurement accuracy: 0.5%

ESP-WROOM-32

1. Overview

ESP32-WROOM-32 is a powerful, generic Wi-Fi+BT+BLE MCU module that targets a wide variety of applications, ranging from low-power sensor networks to the most demanding tasks, such as voice encoding, music streaming and MP3 decoding.

At the core of this module is the ESP32-D0WDQ6 chip*. The chip embedded is designed to be scalable and adaptive. There are two CPU cores that can be individually controlled, and the CPU clock frequency is adjustable from 80 MHz to 240 MHz. The user may also power off the CPU and make use of the low-power co-processor to constantly monitor the peripherals for changes or crossing of thresholds. ESP32 integrates a rich set of peripherals, ranging from capacitive touch sensors, Hall sensors, SD card interface, Ethernet, high-speed SPI, UART, I²S and I²C.

Note:
* For details on the part numbers of the ESP32 family of chips, please refer to the document *ESP32 Datasheet*.

The integration of Bluetooth, Bluetooth LE and Wi-Fi ensures that a wide range of applications can be targeted, and that the module is all-around: using Wi-Fi allows a large physical range and direct connection to the Internet through a Wi-Fi router, while using Bluetooth allows the user to conveniently connect to the phone or broadcast low energy beacons for its detection. The sleep current of the ESP32 chip is less than 5 μA, making it suitable for battery powered and wearable electronics applications. The module supports a data rate of up to 150 Mbps, and 20 dBm output power at the antenna to ensure the widest physical range. As such the module does offer industry-leading specifications and the best performance for electronic integration, range, power consumption, and connectivity.

The operating system chosen for ESP32 is freeRTOS with LwIP; TLS 1.2 with hardware acceleration is built in as well. Secure (encrypted) over the air (OTA) upgrade is also supported, so that users can upgrade their products even after their release, at minimum cost and effort.

Table 1 provides the specifications of ESP32-WROOM-32.

Table 1: ESP32-WROOM-32 Specifications

Categories	Items	Specifications
Certification	RF certification	FCC/CE-RED/IC/TELEC/KCC/SRRC/NCC
	Wi-Fi certification	Wi-Fi Alliance
	Bluetooth certification	BQB
	Green certification	RoHS/REACH
Test	Reliablity	HTOL/HTSL/uHAST/TCT/ESD
Wi-Fi	Protocols	802.11 b/g/n (802.11n up to 150 Mbps)
		A-MPDU and A-MSDU aggregation and 0.4 μs guard interval support
	Frequency range	2.4 GHz ~ 2.5 GHz
Bluetooth	Protocols	Bluetooth v4.2 BR/EDR and BLE specification
	Radio	NZIF receiver with –97 dBm sensitivity
		Class-1, class-2 and class-3 transmitter
		AFH

POWER SUPPLY

1. Mini-ultra Power Supply Module

The 20W ultra-small series power supply module is a small volume, high efficiency power module for customers designed by Hi-Link. It has the advantage of global input voltage range, low temperature rise, low power consumption, high efficiency, high reliability, high security isolation etc. and has been widely used in smart home, automation control, communication equipment, instruments and other industries.

2. Product Model

Model	Dimension (mm)	Output power (W)	Output Voltage (V)	Output Current (mA)	Notes
HLK-20M05	56*32*22.5	20	5	4000	
HLK-20M09			9	2200	
HLK-20M12			12	1600	
HLK-20M15			15	1300	
HLK-20M24			24	830	

3. Product Characteristic

1. Ultra thin, ultra small, minimum volume in the industry
2. Universal input voltage (90~265Vac)
3. Low power consumption, environmental protection, no-load loss<0.1W
4. Low ripple and low noise
5. Good output short circuit, over-current protection and self-recovery
6. High efficiency and high power density
7. Input-output isolated voltage-proof 3000Vac
8. 100% full load aging and testing
9. High reliability, long life design, continuous working time more than 100000 hours
10. Meet UL, CE requirements; product design meets EMC and safety test requirements
11. Adopt high quality environmental protection waterproof heat conduction glue to fill seal, moisture-proof, anti-vibration, meet the IP65 standard of waterproof and dust proof
12. Economic solution, cost-effective
13. Working without an external circuit
14. 1 year warranty

Rev 2; 6/05

DALLAS SEMICONDUCTOR MAXIM

Extremely Accurate I²C-Integrated RTC/TCXO/Crystal

DS3231

General Description

The DS3231 is a low-cost, extremely accurate I²C real-time clock (RTC) with an integrated temperature-compensated crystal oscillator (TCXO) and crystal. The device incorporates a battery input, and maintains accurate timekeeping when main power to the device is interrupted. The integration of the crystal resonator enhances the long-term accuracy of the device as well as reduces the piece-part count in a manufacturing line. The DS3231 is available in commercial and industrial temperature ranges, and is offered in a 16-pin, 300-mil SO package.

The RTC maintains seconds, minutes, hours, day, date, month, and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator. Two programmable time-of-day alarms and a programmable square-wave output are provided. Address and data are transferred serially through an I²C bidirectional bus.

A precision temperature-compensated voltage reference and comparator circuit monitors the status of V_{CC} to detect power failures, to provide a reset output, and to automatically switch to the backup supply when necessary. Additionally, the $\overline{RST}$ pin is monitored as a pushbutton input for generating a reset externally.

Applications

Servers
Utility Power Meters
Telematics
GPS

Pin Configuration appears at end of data sheet.

Features

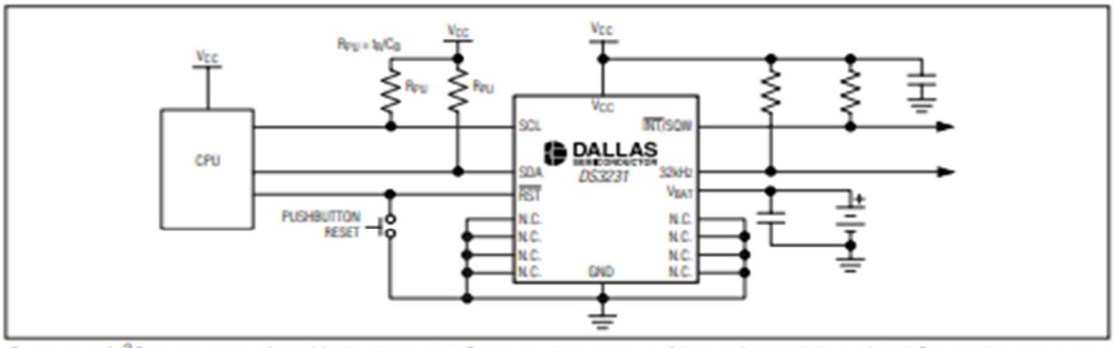
- Accuracy ±2ppm from 0°C to +40°C
- Accuracy ±3.5ppm from -40°C to +85°C
- Battery Backup Input for Continuous Timekeeping
- Operating Temperature Ranges
  Commercial: 0°C to +70°C
  Industrial: -40°C to +85°C
- Low-Power Consumption
- Real-Time Clock Counts Seconds, Minutes, Hours, Day, Date, Month, and Year with Leap Year Compensation Valid Up to 2100
- Two Time-of-Day Alarms
- Programmable Square-Wave Output
- Fast (400kHz) I²C Interface
- 3.3V Operation
- Digital Temp Sensor Output: ±3°C Accuracy
- Register for Aging Trim
- $\overline{RST}$ Input/Output
- UL Recognized

Ordering Information

PART	TEMP RANGE	PIN-PACKAGE	TOP MARK
DS3231S	0°C to +70°C	16 SO	DS3231
DS3231SN	-40°C to +85°C	16 SO	DS3231N
DS3231S+	0°C to +70°C	16 SO	DS3231+
DS3231SN+	-40°C to +85°C	16 SO	DS3231N+

+Denotes lead-free

Typical Operating Circuit



Purchase of I²C components from Maxim Integrated Products, Inc., or one of its sublicensed Associated Companies, conveys a license under the Philips I²C Patent Rights to use these components in an I²C system, provided that the system conforms to the I²C Standard Specification as defined by Philips.

For pricing, delivery, and ordering information, please contact Maxim/Dallas Direct! at 1-888-629-4642, or visit Maxim's website at www.maxim-ic.com.

A1220, A1221, and A1222

Chopper Stabilized Precision Hall Effect Latches

Features and Benefits

- Symmetrical latch switchpoints
- Resistant to physical stress
- Superior temperature stability
- Output short-circuit protection
- Operation from unregulated supply down to 3 V
- Reverse battery protection
- Solid-state reliability
- Small package sizes

Packages:



3-pin SOT23W (suffix LH)



3-pin SIP (suffix UA)

Not to scale

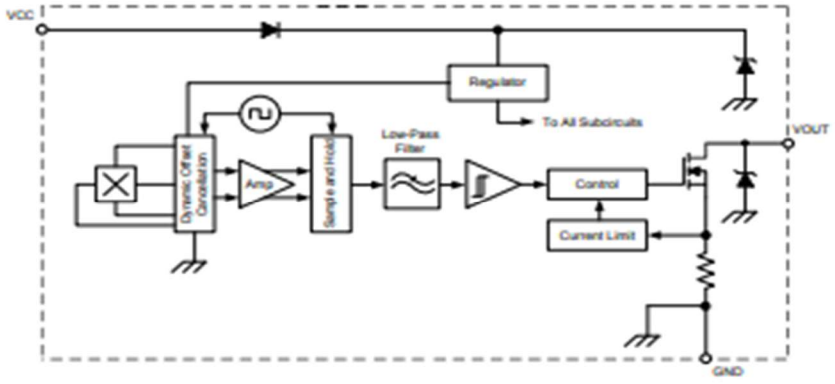
Description

The A1220, A1221, and A1222 Hall-effect sensor ICs are extremely temperature-stable and stress-resistant devices especially suited for operation over extended temperature ranges to 150°C. Superior high-temperature performance is made possible through dynamic offset cancellation, which reduces the residual offset voltage normally caused by device overmolding, temperature dependencies, and thermal stress. Each device includes on a single silicon chip a voltage regulator, Hall-voltage generator, small-signal amplifier, chopper stabilization, Schmitt trigger, and a short-circuit protected open-collector output to sink up to 25 mA. A south pole of sufficient strength turns the output on. A north pole of sufficient strength is necessary to turn the output off.

An onboard regulator permits operation with supply voltages of 3 to 24 V. The advantage of operating down to 3 V is that the device can be used in 3-V applications or with additional external resistance in series with the supply pin for greater protection against high voltage transient events.

Two package styles provide magnetically optimized packages for most applications. Package type LH is a modified 3-pin SOT23W surface mount package while UA is a three-pin ultra-mini SIP for through hole mounting. Both packages are lead (Pb) free, with 100% matte tin plated leadframes.

Functional Block Diagram

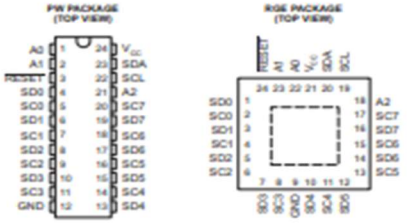


LOW VOLTAGE 8-CHANNEL I²C SWITCH WITH RESET

Check for Samples: TCA9548A

FEATURES

- 1-of-8 Bidirectional Translating Switches
- I²C Bus and SMBus Compatible
- Active-Low Reset Input
- Address by Three Hardware Address Pins for Use of up to Eight Devices
- Channel Selection Via I²C Bus
- Power-Up with All Switch Channels Deselected
- Low r_{ON} Switches
- Allows Voltage-Level Translation Between 2.5-V, 3.3-V, and 5-V Buses
- No Glitch on Power Up
- Supports Hot Insertion
- Low Standby Current
- Operating Power-Supply Voltage Range of 1.65 V to 5.5 V
- 5-V-Tolerant Inputs
- 400-kHz Fast I²C Bus
- Latch-Up Performance Exceeds 100 mA Per JESD 78, Class II
- ESD Protection Exceeds JESD 22
  - 2000-V Human-Body Model (A114-A)
  - 200-V Machine Model (A115-A)
  - 1000-V Charged-Device Model (C101)



DESCRIPTION/ORDERING INFORMATION

The TCA9548A has eight bidirectional translating switches that can be controlled via the I²C bus. The SCL/SDA upstream pair fans out to eight downstream pairs, or channels. Any individual SCx/SDx channel or combination of channels can be selected, determined by the contents of the programmable control register.

The system master can reset the TCA9548A in the event of a timeout or other improper operation by asserting a low in the $\overline{\text{RESET}}$ input. Similarly, the power-on reset deselects all channels and initializes the I²C/SMBus state machine. Asserting $\overline{\text{RESET}}$ causes the same reset/initialization to occur without powering down the part.

The pass gates of the switches are constructed so that the V_{CC} pin can be used to limit the maximum high voltage, which is passed by the TCA9548A. This allows the use of different bus voltages on each pair, so that 1.8-V or 2.5-V or 3.3-V parts can communicate with 5-V parts, without any additional protection. External pullup resistors pull the bus up to the desired voltage level for each channel. All I/O pins are 5-V tolerant.

ORDERING INFORMATION

T_A	PACKAGE(1) (2)		ORDERABLE PART NUMBER	TOP-SIDE MARKING
–40°C to 85°C	QFN – RGE	Reel of 3000	TCA9548ARGER	PW548A
	TSSOP – PW	Reel of 2000	TCA9548APWR	
		Tube of 60	TCA9548APW	

(1) For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI website at www.ti.com.
(2) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

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PRODUCTION DATA information is current as of publication date.

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